

SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that the regular hexagon of side length one cannot be covered by seven open equilateral triangles of side length one. The center and the six vertices canonically label the seven covering triangles. A midpoint-supplier reduction routes every hypothetical cover to one of three strategies: trace-length deficit, area loss, or a finite equilateral-enclosure obstruction. The finite-enclosure strategy uses a fixed six-point selected-gap witness for either nonzero gap rank, an unchanged four-point supported-rescuer witness, and the explicit zero-gap nine-point theorem. A direct neighboring-ray bound removes V-type splits along nonsupercritical paths. Appendices collect the exact local reach calculations certifying that the relevant finite sets do not fit in a unit C triangle. The paper is organized as one proof, and the two disk-tangent contact bounds are verified in a self-contained exact certificate appendix over $\mathbb{Q}(\sqrt{3})$.

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1. INTRODUCTION AND PROOF OUTLINE

1.1. The theorem and the canonical labeling. Let

$$O = (0, 0), \quad V_i = \left(\cos \frac{i\pi}{3}, \sin \frac{i\pi}{3} \right), \quad H = \text{conv}\{V_0, \dots, V_5\},$$

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where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \quad r_i = [O, V_i], \quad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^5 r_i.$$

Then

$$(1) \quad \begin{aligned} \mathcal{H}^1(\partial H) &= 6, \\ \mathcal{H}^1\left(\bigcup_{i=0}^5 r_i\right) &= 6, \\ \mathcal{H}^1(S) &= 12. \end{aligned}$$

Theorem 1.1 (Main theorem). *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem). *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

The seven points $O, V_0, \dots, V_5$ are pairwise at distance at least one, whereas any two points in an open unit equilateral triangle are at distance strictly less than one. Hence the seven covering triangles admit a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \quad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle* at V_i . Set

$$T_C = \overline{U_C}, \quad T_i = \overline{U_i}.$$

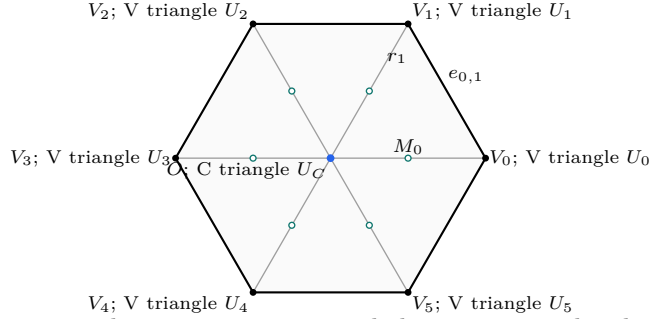


FIGURE 1. The center, vertices, radial segments, and radial mid-points of H .

1.2. Finite classifications. Set

$$N_E(T_C) := |\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\}|.$$

The only possible values are 0, 1, 2. We call the corresponding C triangle types CE0, CE1, CE2; thus

$$T_C \text{ has type CE0} \iff N_E(T_C) = 0,$$

$$T_C \text{ has type CE1} \iff N_E(T_C) = 1,$$

$$T_C \text{ has type CE2} \iff N_E(T_C) = 2.$$

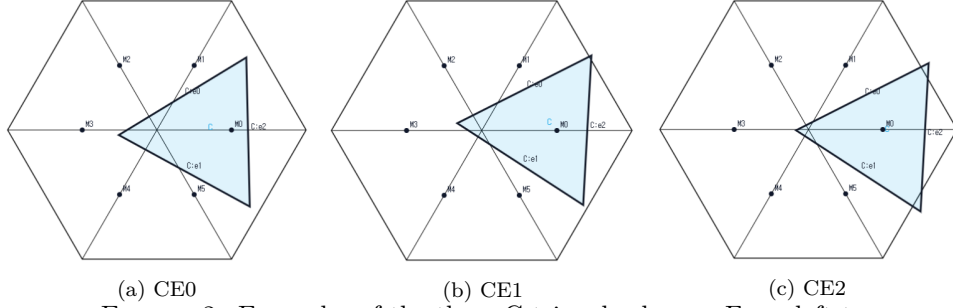


FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H ; point contacts do not count. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

For the V triangle T_i , let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = |\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\}|,$$

with indices modulo 6. The raw pair (3,0) says that all three vertices of T_i lie outside H ; it does not say that T_i is disjoint from H , since $V_i \in \text{int}(T_i) \cap H$. Proposition 2.5 replaces every such triangle by a same-orientation translate with exactly the same trace in H , still containing V_i in its interior, and with at most two vertices outside H . We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
(1,0)	Vd0
(2,0)	Vd0
(1,1)	Vd1
(1,2)	Vd2
(2,1)	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0, 1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$

$$B_i := \max\{x \in [0, 1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$

$$C_i := \max\{x \in [0, 1] : V_i + x(O - V_i) \in T_i\}.$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i . The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and

$$(2) \quad N_+ := |\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}|.$$

Lowercase reaches are used only after a selection satisfying

$$0 \leq a_i \leq A_i,$$

$$0 \leq b_i \leq B_i,$$

$$0 \leq c_i \leq C_i.$$

They never define N_+ .

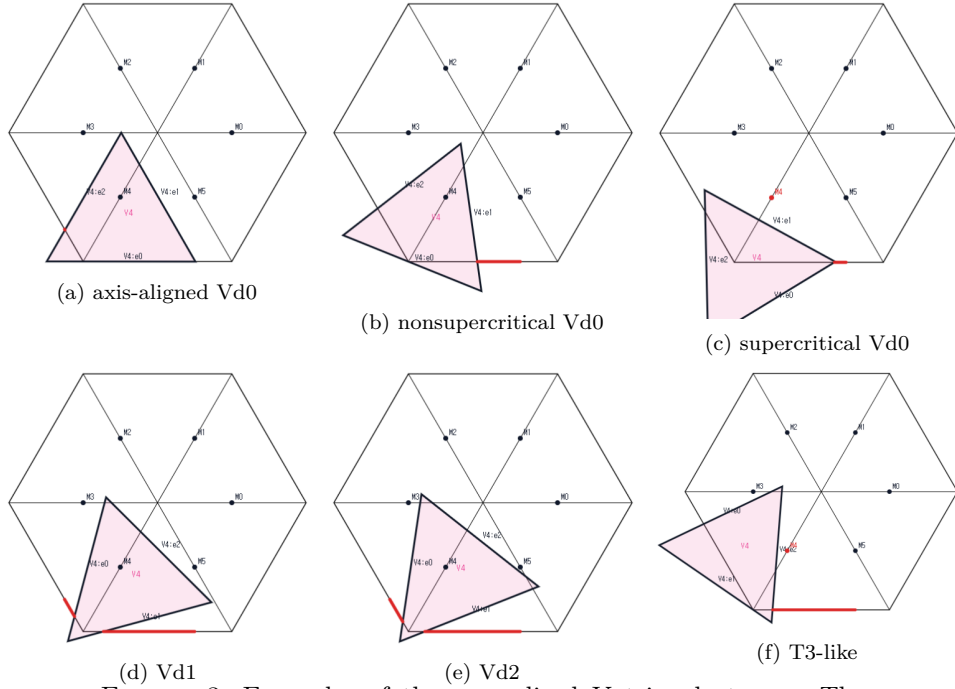


FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \leq 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The screenshots are illustrative and not to scale; no proof depends on visual inspection.

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \quad 0 \leq x \leq 1.$$

The incident open traces are

$$U_i \cap e_{i,i+1} = X_i([0, B_i)),$$

$$U_{i+1} \cap e_{i,i+1} = X_i((1 - A_{i+1}, 1]).$$

Define the *boundary gap*

$$J_i := e_{i,i+1} \setminus ((U_i \cap e_{i,i+1}) \cup (U_{i+1} \cap e_{i,i+1})).$$

The displayed trace identities give

$$J_i \neq \emptyset \iff B_i + A_{i+1} \leq 1.$$

Equality gives a singleton gap, which remains uncovered. Put

$$N_{\text{gap}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|.$$

For the case tables, put

$$d := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}\}|,$$

$$t := |\{i \in \mathbb{Z}/6\mathbb{Z} : (o(T_i), n(T_i)) = (2, 1)\}|,$$

and

$$N_{\text{sp}} := |\{i \in \mathbb{Z}/6\mathbb{Z} : n(T_i) > 0\}| = d + t.$$

The common trace calculation proves that the number of V triangles that are either supercritical or meet an adjacent radial segment in positive length is

$$N_+ + N_{\text{sp}}.$$

In particular, every CE1/CE2 configuration with $N_+ + N_{\text{sp}} \geq 3$ is impossible by a trace-length bound on S .

1.3. Exhaustive routing. Zero actual gaps mean that the six open V triangles cover ∂H . The zero-gap route depends only on N_+ , not the C or V types. For nonzero gaps, N0 and the skeleton budget first reduce to one supercritical role and at most one positive-support role. The remaining distinction is whether the supercritical role or its midpoint supplier is based at the C midpoint. This replaces a separate case tree for each V-type count and each CE class.

N_{gap}	N_+	Additional condition	Closing mechanism
0	0	Arbitrary V types	Strict overlap
0	1	Arbitrary V types	Nine-point F
0	≥ 2	Arbitrary V types	Cyclic area loss
≥ 1	0	Arbitrary V types	N0, through BC
≥ 1	≥ 1	$N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$	Skeleton budget
≥ 1	1	$N_{\text{sp}} \leq 1$	Midpoint supplier: BC, D, replacement, or Vd2 length

TABLE 1. Disjoint routes for the original open cover. Nonzero gaps force CE1 or CE2; both classes share the same placement reduction. Replacement preserves the skeleton and ends at N0, without preserving gap rank.

The exact structural normalization is theorem 2.15. The last row is resolved by theorem 5.16; the local CE1 and CE2 calculations remain separate inside the BC path theorem, but the gap count no longer changes the point construction. Optional A and the four-point B refinement are retained in Appendix D, outside the minimum proof route.

1.4. Organization of the proof. Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the finite residual-hull principle and the nonzero-gap and zero-gap enclosure obstructions. Section 6 then applies the zero-gap alternatives before the nonzero-gap midpoint-supplier reduction of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. COMMON GEOMETRY AND STRUCTURAL REDUCTIONS

The introduction fixes the exact C triangle and V triangle classifications, the actual maximal reaches A_i, B_i, C_i , the selected lower bounds a_i, b_i, c_i , and the boundary gaps J_i . This section proves that those definitions give the finite structural alternatives used by all three mechanisms. Coordinate evaluations needed in the proofs are collected in Appendix A.

2.1. Open and closed formulations.

Proposition 2.1 (Compactness and scaling equivalence). *The following assertions are equivalent.*

- (1) H is covered by seven open equilateral triangles of side 1.
- (2) For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.
- (3) For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side 1.

Lemma 2.2 (Distinct triangles). *The seven triangles can be labelled so that*

$$O \in U_C, \quad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \text{int}(T_C), \quad V_i \in \text{int}(T_i).$$

2.2. C triangle and V triangle classifications. Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.3 (Exhaustive center types). *The closed C triangle T_C has exactly one of the types CE0, CE1, CE2; equivalently,*

$$N_E(T_C) \leq 2.$$

Lemma 2.4 (Local wedge). *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H ; if both adjacent radial traces have positive length, at least two vertices belong to H .*

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.5 (Exact-trace normalization of raw $(3, 0)$ roles). *Let U be an original open V triangle at V_i , put $T = \bar{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\hat{T}$, with closure $\hat{T}$, such that*

$$(3) \quad V_i \in \hat{U}, \quad (o(\hat{T}), n(\hat{T})) = (2, 0),$$

and

$$(4) \quad T \cap H = \widehat{T} \cap H, \quad U \cap H = \widehat{U} \cap H.$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.6 (Exhaustive normalized vertex types). *Every normalized V triangle has exactly one of the four types Vd0, Vd1, Vd2, T3-like in the exact dictionary of the introduction.*

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.7 (T3-like closed-trace domination). *Let T be the closure of an original T3-like V triangle at V_i . There is a same-orientation translate $\widehat{T}$ such that*

$$V_i \in \text{relint}(a \text{ side of } \widehat{T}), \quad T \cap H \subsetneq \widehat{T} \cap H,$$

and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$, while

$$\text{int}(T) \cap (H \setminus \{V_i\}) \subseteq \text{int}(\widehat{T}).$$

Since V_i becomes a boundary point, $\widehat{T}$ is only a closed-trace majorant.

2.3. Initial reaches and boundary gaps. The exact reach definitions are those in the introduction. The following elementary fact records how their closed maxima and open suprema are used.

Lemma 2.8 (Initial reach segments). *On the three segments from V_i toward V_{i-1}, V_{i+1}, O , the closed parameter sets are respectively*

$$[0, A_i], \quad [0, B_i], \quad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \quad [0, B_i), \quad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

The gap J_i is the exact set difference defined in the introduction. The name includes the equality case $B_i + A_{i+1} = 1$: then J_i is a singleton missed by both incident open V triangles.

Lemma 2.9 (Exhaustiveness of the gap split). *Every J_i is empty or one closed interval, with singleton intervals retained. Under a hypothetical cover,*

$$N_{\text{gap}} = |\{i \in \mathbb{Z}/6\mathbb{Z} : J_i \neq \emptyset\}|,$$

and every nonempty J_i lies in a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 permits respectively at most 0, 1, 2 gaps, and

$$N_{\text{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Lemma 2.10 (T3-like V triangles are nonsupercritical). *Every original T3-like V triangle satisfies*

$$(5) \quad A_i + B_i \leq 1.$$

It is therefore not counted by N_+ .

2.4. Strict handoffs and midpoint forcing.

Proposition 2.11 (Strict boundary handoffs). *Assume that $U_0, \dots, U_5$ cover ∂H . There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i . If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Lemma 2.12 (Boundary deficit and overlap identities). *For the actual reaches put $s_i = 1 - A_i - B_i$ and $\omega_i = B_i + A_{i+1} - 1$. Then*

$$A_{i+1} - A_i = s_i + \omega_i, \quad B_i - B_{i+1} = s_{i+1} + \omega_i.$$

On a nonsupercritical gap-free path the A_i increase and the B_i decrease strictly. Weak handoffs give the weak version. Around the full cycle, $\sum_i \omega_i = -\sum_i s_i$.

Lemma 2.13 (Self-midpoint obstruction). *Let u, v be unit vectors meeting at 120° . If a closed unit equilateral triangle contains*

$$0, \quad au, \quad bv, \quad \frac{u+v}{2},$$

then $a + b \leq 1$. Consequently

$$M_i \in T_i \implies A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proposition 2.14 (CE1/CE2 exactly-one-midpoint property). *If $O \in \text{int}(T_C)$ and T_C is CE1 or CE2, then*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1.$$

The interior hypothesis is essential: $\text{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1 , but O is not interior to it.

2.5. Exhaustive structural routing.

Proposition 2.15 (Exhaustive structural reduction). *Every hypothetical cover admits the distinct C triangle and V triangles used in Table 1, and its actual maximal reaches determine exactly one row of that table.*

3. TRACE-LENGTH MECHANISM

For a closed triangle T and a rectifiable target E , define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Lemma 3.1 (Connected open-cover budget). *Let X be compact and connected, with a finite Borel measure μ positive on every nonempty relatively open subset. For a finite open cover with at least two nonempty relative pieces,*

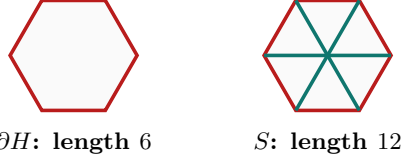
$$\mu(X) < \sum_j \mu(U_j \cap X) \leq \sum_j \mu(T_j \cap X).$$

Consequently closed-piece upper bounds whose sum is at most $\mu(X)$ already contradict the open cover.

For $X = \partial H$ or S , use length; for $X = H$, use area. Each measure has the stated full-support property, and the seven distinguished roles meet the corresponding targets in positive measure. In particular,

$$(6) \quad \mathcal{H}^1(E) < L_E(T_C) + \sum_{i=0}^5 L_E(T_i), \quad E \in \{\partial H, S\},$$

whenever these targets are covered. Empty center boundary traces are simply omitted from the collection. The local geometric caps remain separate from this common final budget argument.



the full-skeleton estimate is used only under its stated branch hypotheses

FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton interfaces.

Lemma 3.2 (Center skeleton cap). *If $N_E(T_C) \in \{1, 2\}$ and*

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Lemma 3.3 (Positive-support skeleton cap). *Let T_i be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.4 (No-support skeleton cap). *If*

$$A_i + B_i \leq 1,$$

$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \quad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Lemma 3.5 (Supercritical skeleton cap). *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Lemma 3.6 (Positive-support rescuer). *Suppose $N_E(T_C) \in \{1, 2\}$ and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \quad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Theorem 3.7 (Direct CE1/CE2 skeleton count). *If*

$$N_E(T_C) \in \{1, 2\},$$

$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

and

$$(7) \quad N_+ + N_{\text{sp}} \geq 3,$$

then the seven triangles do not cover S .

3.2. Boundary trace register. The exact evaluation is in Appendix B; the following table is the complete interface used by the routing argument.

Theorem 3.8 (Boundary trace table). *For closures of original open triangles, the following bounds hold.*

	<i>exact hypothesis</i>	<i>established class</i>	<i>exact bound</i>
	$N_E(T_C) = 0$	CE0	$L_{\partial H}(T_C) = 0$
	$N_E(T_C) = 1$	CE1	$L_{\partial H}(T_C) \leq \frac{\sqrt{3}}{2} - \frac{3}{4}$
	$N_E(T_C) = 2$	CE2	$L_{\partial H}(T_C) < \frac{1}{2}$
(8)	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i \leq 1$	Vd0	$L_{\partial H}(T_i) \leq 1$
	$(o(T_i), n(T_i)) \in \{(1, 0), (2, 0)\}$ $A_i + B_i > 1$	Vd0	$L_{\partial H}(T_i) \leq \frac{2}{\sqrt{3}}$
	$(o(T_i), n(T_i)) \in \{(1, 1), (1, 2)\}$	Vd1 or Vd2	$L_{\partial H}(T_i) < \frac{1}{2}$
	$(o(T_i), n(T_i)) = (2, 1)$	T3-like	$L_{\partial H}(T_i) < 1.$

For every V triangle,

$$(9) \quad L_{\partial H}(T_i) = A_i + B_i.$$

Corollary 3.9 (Branches closed by the common budgets). *The following exact alternatives are impossible.*

- (1) $N_E(T_C) = 0$, $N_+ = 1$, and $d \geq 1$;
- (2) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (3) $N_E(T_C) = 1$, $N_+ = 1$, and $d \geq 1$;
- (4) $N_E(T_C) = 2$, $N_+ = 1$, and $d \geq 2$;
- (5) $N_E(T_C) = 2$, $N_+ = 1$, and there is an index i such that

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset;$$

- (6) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (7) $N_E(T_C) \in \{1, 2\}$ and $N_+ \geq 2$.

Proposition 3.10 (Vd2 adjacent-midpoint branch). *Suppose the C triangle is $CE1$ or $CE2$ and $N_+ = 1$. If a Vd2 role contains an adjacent midpoint, the perimeter is not covered.*

3.3. Routing conclusion. The zero-gap $N_+ = 1$, $d \geq 1$ clause is retained as an independent alternative; the routing table now sends every zero-gap $N_+ = 1$ state to theorem 5.23.

Proposition 3.11 (Branches closed by the trace-length mechanism). *The following routing entries and hybrid subcase are impossible.*

- (1) $N_{\text{gap}} = 0$ and $N_+ = 0$;
- (2) $N_{\text{gap}} = 0$, $N_+ = 1$, and $d \geq 1$;
- (3) $N_E(T_C) \in \{1, 2\}$ and $N_+ + N_{\text{sp}} \geq 3$;
- (4) $N_E(T_C) \in \{1, 2\}$, $N_+ = 0$, and $d \geq 1$;
- (5) $N_E(T_C) = 1$, $N_+ = 1$, and $(d, t) = (1, 0)$;
- (6) the subcase of $N_E(T_C) = 2$, $N_+ = 1$, and $(d, t) = (1, 0)$ in which the unique non-Vd0 triangle T_i satisfies

$$(o(T_i), n(T_i)) = (1, 2)$$

and

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset.$$

4. AREA-LOSS ESTIMATE

The area-loss method records how much of each V triangle must lie outside H . The local estimate applies to every V triangle. A final cyclic argument adds these losses without imposing a center type.

Put $A_\Delta = \sqrt{3}/4$ and normalize all areas by A_Δ . At V_0 define

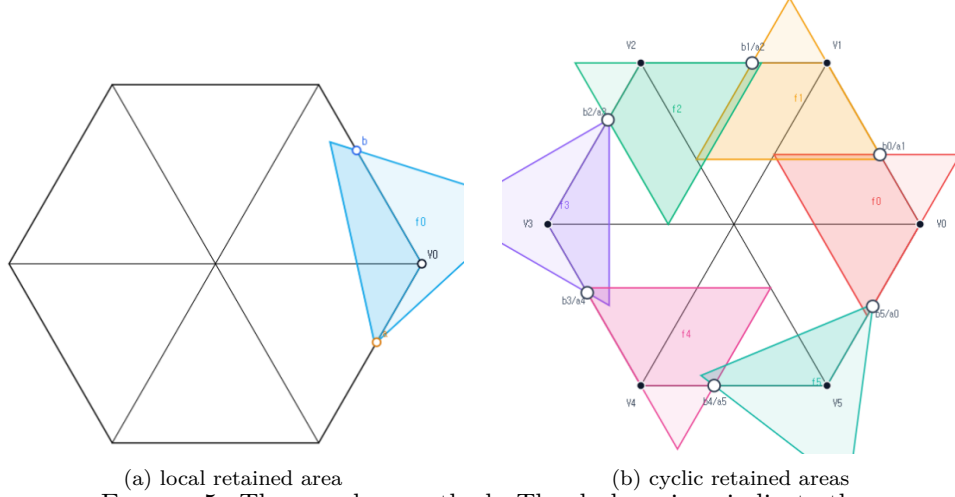
$$\mathcal{T}(a, b) := \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ \{V_0, V_0 + a(V_5 - V_0), V_0 + b(V_1 - V_0)\} \subseteq T \end{array} \right\},$$

$$\mathcal{L}(T) := \frac{\text{area}(T \setminus H)}{A_\Delta}, \quad \mathcal{L}(a, b) := \inf_{T \in \mathcal{T}(a, b)} \mathcal{L}(T),$$

where $a, b \in [0, 1]$ and $\mathcal{T}(a, b) \neq \emptyset$. Hexagon symmetry gives the corresponding family at every V_i . In particular, if

$$0 \leq a_i \leq A_i, \quad 0 \leq b_i \leq B_i,$$

then the rotated copy of T_i belongs to $\mathcal{T}(a_i, b_i)$.



(a) local retained area (b) cyclic retained areas
 FIGURE 5. The area-loss method. The dark regions indicate the portions retained in H ; the exact quantities are $\mathcal{L}(T)$ and $\mathcal{L}(a, b)$.

4.1. Local area loss.

Theorem 4.1 (Local square loss). *For every feasible (a, b) ,*

$$(10) \quad \mathcal{L}(a, b) \geq \min\{a, b\}^2.$$

If $a + b > 1$, then

$$(11) \quad \mathcal{L}(a, b) \geq \max\{a, b\}^2.$$

The same bounds hold for every $T \in \mathcal{T}(a, b)$.

4.2. Cyclic accumulation. Use the strict handoffs from theorem 2.11:

$$\xi_i \in (1 - A_{i+1}, B_i), \quad a_i := 1 - \xi_{i-1}, \quad b_i := \xi_i.$$

They satisfy

$$0 < a_i < A_i, \quad 0 < b_i < B_i, \quad a_i^2 + a_i b_i + b_i^2 < 1.$$

Lemma 4.2 (Cyclic area-loss certificate). *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \implies \sum_{i=0}^5 \mathcal{L}(T_i) > 1.$$

Proposition 4.3 (Branch closed by area). *The branch $N_{\text{gap}} = 0$, $N_+ \geq 2$ is impossible.*

5. FINITE-ENCLOSURE OBSTRUCTIONS

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

For $X \in \mathbb{R}^2$ and $r \geq 0$, put

$$\mathbb{B}(X, r) := \{Y \in \mathbb{R}^2 : \|Y - X\| \leq r\}.$$

For every nonempty compact $K \subseteq \mathbb{R}^2$, define

$$\Lambda(K) := \inf \left\{ s > 0 : \begin{array}{l} \exists T [T \text{ is a closed equilateral triangle of side } s, \\ K \subseteq T] \end{array} \right\}.$$

Thus $\Lambda(K)$ is the least equilateral enclosure side for K .

Lemma 5.1 (Compact set in an open unit triangle). *If K is compact, U is an open unit equilateral triangle, and $K \subset U$, then*

$$\Lambda(K) < 1.$$

5.1. Four-anchor and neighboring-arm capacities. Normalize at V_0 and define the four-anchor hull

$$K(a, b, c) := \text{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \quad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

The exact admissible set, attainment, and the evaluation of these extrema are proved in theorem A.9.

The neighboring-arm capacities are the variational quantities

$$C_+(a, b) := \max \left\{ c \in [0, 1] : \begin{array}{l} \exists T [T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T] \end{array} \right\},$$

$$C_-(a, b) := C_+(b, a), \quad 0 \leq a, b \leq 1, \quad a + b \leq 1.$$

Only a bound, not the full piecewise neighboring capacity, is needed below. The direct four-caliper proof is theorem D.14.

Definition 5.2 (Fixed total radial endpoint). Parametrize r_i by $Z_i(c) = (1-c)V_i$, measured from V_i toward O . Let $u_{j \rightarrow i}$ be the largest parameter in the actual positive trace of T_j on r_i for $j = i-1, i+1$, with value zero when no such positive trace exists. Put

$$\gamma_i := \max\{C_i, u_{i-1 \rightarrow i}, u_{i+1 \rightarrow i}\}, \quad d_i := 1 - \gamma_i, \quad \hat{P}_i := d_i V_i.$$

Thus $\hat{P}_i$ is the innermost endpoint reached by any local V triangle on r_i , rather than just the endpoint of the own V triangle. This distinction is essential when a nonsupercritical V triangle supports an adjacent radial segment.

Lemma 5.3 (Fixed radial forcing and demand recovery). *One has $0 < d_i < 1$ and $\hat{P}_i \notin \bigcup_j U_j$. Under skeleton coverage, $\hat{P}_i \in U_C$. More generally, for every $\gamma_i \leq \Gamma \leq 1$, the point $(1-\Gamma)V_i$ is missed by every open V triangle and is center-forced under skeleton coverage. If an arbitrary open unit candidate U' contains*

O and $\widehat{P}_i$, and its exit on r_i is d'_i measured from O , then $\gamma_i > 1 - d'_i$. If both neighboring endpoints are at most $1 - d'_i$, it follows that

$$C_i > 1 - d'_i.$$

It is sufficient that each neighboring V triangle be nonsupercritical with both actual boundary reaches greater than d'_i .

Let selected lower bounds satisfy

$$\begin{aligned} 0 \leq a_j \leq A_j, \quad 0 \leq b_j \leq B_j \quad (0 \leq j \leq 5), \\ a_j^2 + a_j b_j + b_j^2 \leq 1 \quad (0 \leq j \leq 5). \end{aligned}$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\begin{aligned} \mathcal{H}^1(T_{i-1} \cap r_i) > 0 &\implies a_{i-1} + b_{i-1} \leq 1, \\ \mathcal{H}^1(T_{i+1} \cap r_i) > 0 &\implies a_{i+1} + b_{i+1} \leq 1. \end{aligned}$$

Define

$$\begin{aligned} \Gamma_i := \max \Big(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \\ \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\} \Big), \quad D_i := (1 - \Gamma_i)V_i. \end{aligned}$$

The points D_i are the radial witnesses used below.

Lemma 5.4 (Type-aware radial forcing). *For every i ,*

$$D_i \in r_i \subseteq S, \quad D_i \notin \bigcup_{j=0}^5 U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^5 U_j \implies D_i \in U_C.$$

Lemma 5.5 (Common-pair domination). *If*

$$p, q \geq 0, \quad p + q \leq 1,$$

then

$$C_+(p, q), C_-(p, q) \leq 1 - \min\{p, q\} \leq c_{\max}(p, q).$$

Corollary 5.6 (Uniform common-pair radial forcing). *Suppose $p, q \geq 0$, $p + q \leq 1$, and*

$$p \leq a_i \leq A_i, \quad q \leq b_i \leq B_i \quad (0 \leq i \leq 5), \quad 0 < c_* := c_{\max}(p, q) < 1.$$

Then $(1 - c_)V_i \notin \bigcup_{j=0}^5 U_j$ for every i , without a restriction on the normalized V types. Under a hypothetical cover of H ,*

$$(1 - c_*)V_i \in U_C \quad (0 \leq i \leq 5), \quad \mathbb{B}(O, h(1 - c_*)) \subset U_C.$$

Lemma 5.7 (Open trace endpoint). *Let U be an open equilateral triangle and $T = \overline{U}$. If $T \cap [V, O]$ is the closed interval from V to P , then $P \notin U$. If a noncentral point $Q \in [V, O]$ lies in U , then $\mathcal{H}^1(U \cap [V, O]) > 0$.*

5.2. Fixed witnesses and the case register. An obstruction for the original C triangle is not, by itself, an enclosure inequality. We therefore separate two assertions: the original cover forces one fixed finite set into U_C , and every open unit equilateral candidate fails to contain that same set. Candidate center parameters are recomputed in the second assertion; the witness points are never recomputed.

The actual gap on $e_{i,i+1}$ is

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

Both endpoints are missed by the incident open V triangles and by all nonincident V triangles. Equality $B_i = 1 - A_{i+1}$ is a singleton gap. Convexity fills a gap from its endpoints, so a candidate containing the endpoints of all actual gaps completes the boundary with the original six V triangles. This observation does not assert that it completes the whole skeleton.

Lemma 5.8 (The origin is implicit in the path witnesses). *For $G = X_0(t)$, $0 < t < 1$, and $d_2, d_4 > 0$,*

$$G + \frac{1-t}{d_2}(d_2V_2) + \frac{1}{d_4}(d_4V_4) = 0.$$

Consequently $O \in \text{conv}\{G, d_2V_2, d_4V_4\}$.

There are three active witness recipes and three principal mechanisms: a non-supercritical boundary path, a supported rescuer, and the zero-gap asymmetric construction. The complementary-gap disk is retained in Appendix D as an independent alternative A; the former four-point B proof is also retained there as an optional refinement. Neither is a dependency of the all-nonsupercritical theorem. In the nonzero-gap rows normalize the C triangle's distinguished radial midpoint to M_0 . When $N_+ = 1$, let σ be the unique supercritical index. All point counts below are upper bounds; coincident gap endpoints can reduce them.

Table 2: Fixed finite-enclosure recipes. Ordinary nonsupercritical V triangles are not subdivided by type.

Conditions	Fixed witness set	Disk and count
BC: at least one incident gap; $T_1, \dots, T_5$ nonsupercritical	$\{M_0, X_0(B_0), X_0(1 - A_1), \hat{P}_2, \hat{P}_3, \hat{P}_4\}$	No disk; at most 6
D: center-based T3-like or Vd1 rescuer; one or two gaps	$\{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\}$	No disk; at most 4
F: zero gaps, $N_+ = 1$	Six common radial points Q_-, Q_0, Q_+	Disk; at most 9

Replacement is a skeleton-preserving operation, not a fourth witness recipe. For $N_+ = 1$ and $\sigma = 0$, select either actual gap and use BC. The two boundary witnesses come only from that selected gap. This includes both adjacent and nonadjacent Vd placements: their location no longer changes the active point construction. When $\sigma \neq 0$, midpoint rescue and the retained length filters give the D or R placements specified below. A Vd2 rescuer retains its perimeter exit; the four-point D theorem

does not assume endpoint inequalities that have only been proved for T3-like and Vd1 triangles.

5.3. BC: one selected gap and one six-point construction. Normalize the center midpoint to M_0 and choose any actual gap, reflected to J_0 . The other incident edge $e_{5,0}$ may or may not contain a gap. Only the four middle edges $e_{1,2}, \dots, e_{4,5}$ must be gap-free. The following boundary transfer supplies precisely the input that the old one-gap proof obtained from its fifth handoff.

Lemma 5.9 (Shared-gap-anchor boundary transfer). *If the original open roles cover ∂H and J_0 is an actual gap, then, independently of the gap status of $e_{5,0}$,*

$$B_5 > 1 - M_0(B_0) > B_0/2.$$

Here $M_0(z)$ is the scalar diameter envelope.

In the pure enclosure statement below the tail bound $B_5 \geq B_0/2$ is explicit. In the covering application, the preceding lemma establishes it for the original roles before an arbitrary enclosing candidate is chosen. We do not assume that the candidate also covers the discarded gap.

Define

$$K_{BC} := \{M_0, X_0(B_0), X_0(1 - A_1), \widehat{P}_2, \widehat{P}_3, \widehat{P}_4\}.$$

The midpoint is a structural anchor belonging to the original open C triangle. The other points are missed by the V triangles. The origin is implicit by theorem 5.8; the construction uses at most six points even though a candidate must also contain O .

Lemma 5.10 (The fixed transverse witness is center-forced). *Under the selected-gap hypotheses, with the original C triangle's distinguished midpoint M_0 , skeleton coverage implies $K_{BC} \subset U_C$.*

Proposition 5.11 (CE2 transverse threshold terminal). *Under the BC path hypotheses below, no open unit CE2 candidate contains K_{BC} .*

Proposition 5.12 (CE1 transverse return with neighboring support). *Under the BC path hypotheses below, no open unit CE1 candidate contains K_{BC} .*

Theorem 5.13 (Unified six-point selected-gap enclosure). *Suppose J_0 is an actual gap, $A_i + B_i \leq 1$ for $1 \leq i \leq 5$, and*

$$B_i + A_{i+1} > 1 \quad (1 \leq i \leq 4), \quad B_5 \geq B_0/2.$$

Then $\Lambda(K_{BC}) \geq 1$. There is no condition on the gap status of $e_{5,0}$ and no supercriticality requirement on T_0 . Nonsupercritical path roles are not subdivided by V type. No disk is used.

Theorem 5.14 (Center-aligned path obstruction). *Suppose a skeleton cover has at least one actual gap and its C triangle's unique midpoint is M_k . If every T_i with $i \neq k$ is nonsupercritical, the configuration is impossible.*

Theorem 5.15 (Every skeleton cover has a supercritical V triangle). *A skeleton cover by the seven original open roles satisfies $N_+ \geq 1$.*

Lemma 5.16 (A unique midpoint supplier). *In a nonzero-gap skeleton cover, the preceding N0 theorem and the skeleton budget leave only $N_+ = 1$, $N_{sp} \leq 1$. Write*

M_k for the C midpoint and σ for the supercritical index. If $\sigma \neq k$, there is a unique positive-support role T_τ , where

$$\tau \in \{\sigma - 1, \sigma + 1\}, \quad M_\sigma \in U_\tau.$$

If T_τ is T3-like, then $\tau = k$. If $\tau \neq k$, the edge shared by T_τ, T_σ is center-free, in both CE1 and CE2.

5.4. D: a four-point supported-rescuer enclosure. Let $Y(t) = (1 - t)V_0 + tV_5 = X_5(1 - t)$. The geometric lemma below is stated independently of any V type or covering configuration.

Theorem 5.17 (Four-point rescuer geometry). *Suppose $a \geq 0$, $\varepsilon > 0$, $\beta \geq 0$, and*

$$s = a + \varepsilon \leq 1, \quad a \leq \varepsilon, \quad \beta \leq \frac{\varepsilon}{s}.$$

Then

$$\Lambda\{O, \varepsilon V_1, Y(a), Y(1 - \beta)\} \geq 1.$$

Lemma 5.18 (Scalar rescuer ratio test). *For $0 \leq x, c \leq 1/2$ and $M = (c + \sqrt{c^2 - 8c + 4})/2$,*

$$x \leq 1 - M \iff x^2 + (c - 2)x + c \geq 0.$$

For the two actual adapters normalize T_0 as the rescuer of M_1 , T_1 as uniquely supercritical, and $T_2, \dots, T_5$ as nonsupercritical ordinary roles on the four center-free edges. Write

$$T_0 \cap r_1 = \{(1 - t)V_1 : c \leq t \leq u\}, \quad 0 < c \leq \frac{1}{2} \leq u < 1, \quad \varepsilon = 1 - u.$$

Let $M = M_c^{\text{sup}} = (c + \sqrt{c^2 - 8c + 4})/2$, with the value $1/2$ at $c = 1/2$ understood as the continuous endpoint convention. The local requirements are

$$\varepsilon V_1 \in U_C, \quad C_1 \geq c, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

The separate inequality $A_0 \leq 1 - M$ follows from the last two and is not an additional hypothesis.

The strict-supercritical envelope gives $B_1 < M$; the four ordinary handoffs give $B_5 \leq B_1$. Thus $B_5 < 1 - A_0$, and the two left-gap endpoints are $Y(A_0)$ and $Y(1 - B_5)$. The fixed set is

$$K_D = \{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\}.$$

It lies in U_C and satisfies the four-point theorem with geometric parameters $a = A_0$, $\beta = B_5$. No sum greater than four is needed. There is no disk, and no distinction between one and two actual gaps in this enclosure step.

T3-like local input. In its translated chart put

$$1 \leq D_T \leq 2/\sqrt{3}, \quad R_T = (D_T + \sqrt{4 - 3D_T^2})/2.$$

The actual supported endpoints are

$$c = \frac{D_T(1 + A_0) - 1}{R_T}, \quad u = 1 - \frac{R_T}{D_T} + A_0, \quad \varepsilon = \frac{R_T}{D_T} - A_0.$$

The T3-like chart calculation verifies the local requirements. In particular, with $x = A_0 D_T / R_T$ and $c = x + \theta$, the ratio test becomes the single quadratic $2x^2 + (\theta - 1)x + \theta \geq 0$.

Vd1 local input. With $t \geq 1$ and $d_t = \sqrt{t^2 + t + 1}$, the actual corner data give

$$c = \frac{t(1 - B_0)}{t + 1}, \quad u = \frac{d_t - A_0 - tB_0 - 1}{t}, \quad \varepsilon = \frac{t + 1 - d_t + A_0 + tB_0}{t}.$$

The Vd1 chart calculation verifies the same requirements for CE1 and CE2; it uses no CE2-only center inequality. These two chart calculations remain in Appendix D; their global ending is identical.

Proposition 5.19 (T3-like supported-rescuer terminal). *Suppose $N_+ = 1$, there is exactly one T3-like role and no Vd1/Vd2 role, and there is at least one actual gap. Then the skeleton is not covered.*

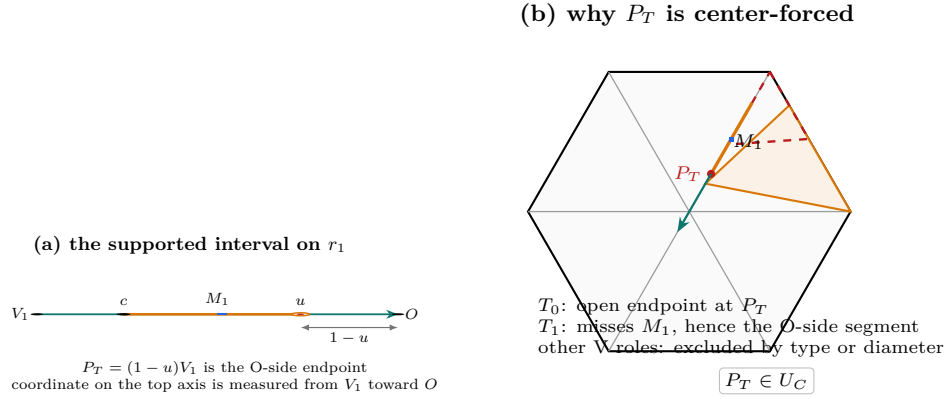


FIGURE 6. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O , so its O-side endpoint is $P_T = (1-u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

5.5. The remaining placement router.

Proposition 5.20 (One-Vd placement assembly). *Suppose the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is Vd1 or Vd2, and there is an actual gap. The skeleton is not covered.*

Proposition 5.21 (All nonzero-gap finite-enclosure rows). *Every nonzero-gap row of table 1 assigned to finite enclosure is impossible.*

5.6. The zero-gap nine-point obstruction. Assume

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

No normalized V-type restriction is imposed. The same nine points work when Vd1, Vd2, or T3-like roles meet adjacent radial segments. Rotate the unique supercritical role to index 4. The exact-one handoff calculation theorem E.2 gives

$$(12) \quad \xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

Define

$$(13) \quad a := 1 - \xi_3, \quad b := \xi_4, \quad p := 1 - b, \quad q := 1 - a.$$

Then

$$(14) \quad \begin{aligned} 0 < a, b < 1, \quad a = a_4 < A_4, \quad b = b_4 < B_4, \\ a + b > 1, \quad a^2 + ab + b^2 < 1, \\ a_i \geq p, \quad b_i \geq q \quad (0 \leq i \leq 5). \end{aligned}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \quad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a, b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p, q), \quad r_* := h(1 - c_*), \quad P_i^{\text{rad}} := (1 - c_*)V_i \quad (0 \leq i \leq 5).$$

The bounds in theorem E.3 give $0 < c_* < 1$. Applying theorem 5.6 to the common pair gives

$$(15) \quad \mathbb{B}(O, r_*) \subseteq \text{conv}\{P_0^{\text{rad}}, \dots, P_5^{\text{rad}}\} \subset U_G.$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \quad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1 - a), 1).$$

Define ℓ_- and ℓ_+ as the corresponding maximal nondegenerate line segments of the interior frontier:

$$\begin{aligned} \{\ell_-\} &:= \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_- \neq \emptyset, \\ \nexists X', Y' [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}, \\ \{\ell_+\} &:= \left\{ [X, Y] : \begin{array}{l} X \neq Y, \quad [X, Y] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H), \\ [X, Y] \cap \mathcal{C}_+ \neq \emptyset, \\ \nexists X', Y' [X, Y] \subsetneq [X', Y'] \subseteq \partial\mathcal{R}_{AB}(a, b) \cap \text{int}(H) \end{array} \right\}. \end{aligned}$$

The three frontier points are the singleton intersections

$$\{Q_0\} := \ell_- \cap \ell_+, \quad \{Q_-\} := \ell_- \cap \mathcal{C}_-, \quad \{Q_+\} := \ell_+ \cap \mathcal{C}_+.$$

Explicit construction of the zero-gap points and disks. One may choose every strict handoff explicitly as

$$\xi_i = \frac{1 - A_{i+1} + B_i}{2}.$$

The exact-one handoff lemma preserves the supercritical index under this choice. At index 4 use the outgoing/incoming chart

$$\Psi_4(u, v) = V_4 + u(V_5 - V_4) + v(V_3 - V_4).$$

Put $\rho = a^2 + ab + b^2$, $\Delta = \sqrt{4\rho - 3}$ and define

$$\begin{aligned} \alpha_F &= \frac{h(b + 2a - b\Delta)}{2\rho}, & \beta_F &= \frac{h(b - a + (a + b)\Delta)}{2\rho}, \\ \gamma_F &= \frac{h(-b + a + (a + b)\Delta)}{2\rho}, & \delta_F &= \frac{h(2b + a - a\Delta)}{2\rho}. \end{aligned}$$

These are frontier coefficients, not the signed C-triangle parameters. The selected first-root parameters are

$$\begin{aligned}\lambda_- &= 2(\alpha_F + b\beta_F) - \frac{2}{3}\sqrt{9(\alpha_F + b\beta_F)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta_F + a\gamma_F) - \frac{2}{3}\sqrt{9(\delta_F + a\gamma_F)^2 - 9a^2 + 9a - 6}.\end{aligned}$$

Then the same three frontier witnesses are

$$\begin{aligned}Q_- &= \Psi_4(b - \beta_F\lambda_-, \alpha_F\lambda_-), & Q_+ &= \Psi_4(\delta_F\mu_+, a - \gamma_F\mu_+), \\ Q_0 &= \Psi_4\left(\frac{2b + a - a\Delta}{\Delta + 3}, \frac{b + 2a - b\Delta}{\Delta + 3}\right).\end{aligned}$$

The exact fixed-line calculation ensures that the radicals are real and selects the stated roots. No sampled point or new root selection is used. The finite set and its compact comparison set are

$$K_F = \{(1 - c_*)V_i : 0 \leq i \leq 5\} \cup \{Q_-, Q_0, Q_+\},$$

$$\mathcal{D}_F = \{x : \|x\| \leq h(1 - c_{\max}(1 - b, 1 - a))\}, \quad \mathcal{D}_F \cup \{Q_-, Q_0, Q_+\} \subseteq \text{conv}(K_F).$$

When $c_* \leq 2/3$, the six radial points already suffice because $\Lambda(\mathcal{D}_F) = 3(1 - c_*) \geq 1$. For $c_* > 2/3$, the unchanged asymmetric enclosure certificate is needed.

Its smaller comparison disk is also explicit. With $p = 1 - b$, $q = 1 - a$, $s = p + q$, $m = \min\{p, q\}$, and $\chi = s^4 - s^2 + pq$, put

$$\bar{c} = \begin{cases} s - \frac{\chi}{4s^3 - 2s + m}, & \chi \leq 0, \\ s - \frac{\chi}{1 - m}, & \chi > 0. \end{cases}$$

Then

$$\overline{\mathcal{D}}_F = \{x : \|x\| \leq h(1 - \bar{c})\} \subseteq \mathcal{D}_F,$$

using the exact bound $c_* \leq \bar{c} < 1$. This smaller disk is a comparison inside the tangent-residual certificate, not an additional global witness recipe. The two unit disks defining the frontier circles are $\mathbb{B}(X_2(b), 1)$ and $\mathbb{B}(X_5(1 - a), 1)$; they are auxiliary distance constructions, not disks asserted to lie in the C triangle.

Their exact construction and exclusions in theorems E.1, E.4 and E.5 use contained vertices and actual handoffs, not Vd0 locality, and give

$$(16) \quad Q_-, Q_0, Q_+ \in U_C.$$

Define the public witness set

$$(17) \quad K_{\text{wit}}(a, b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

Theorem 5.22 (Witness enclosure). *Every pair (a, b) satisfying equation (14) satisfies*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

Theorem 5.23 (Zero-gap obstruction). *There is no hypothetical cover with*

$$N_{\text{gap}} = 0, \quad N_+ = 1.$$

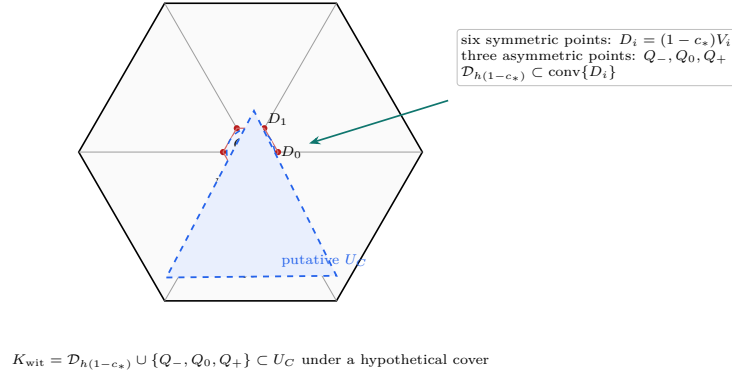


FIGURE 7. The type-independent zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB -frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

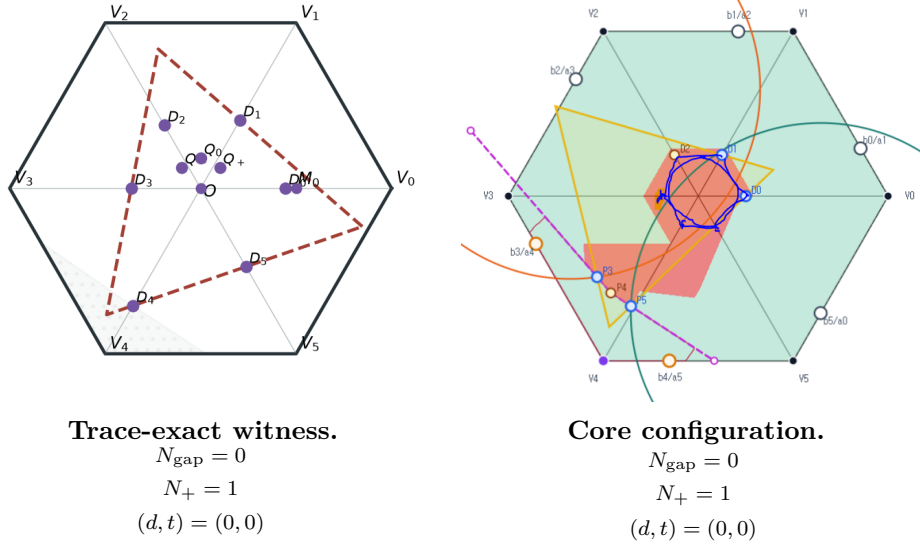


FIGURE 8. Case F: sampled zero-gap obstruction data.

Proposition 5.24 (Cases excluded by the nine-point obstruction). *The entire $N_{\text{gap}} = 0$, $N_+ = 1$ row of table 1 is impossible, independently of the center class and normalized V -type pattern.*

6. COMPLETION OF THE PROOF

The zero-gap certificate uses coverage of all of H . Replacement needs only skeleton coverage and is never used to invoke that certificate. Corollary 1.2 follows from Proposition 2.1.

APPENDIX A. STRUCTURAL, SHARED-LOCAL, AND SIGNED-CENTER OPTIMIZATION

This appendix solves the coordinate and compactness problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. Open–closed shrinking feasibility.

Lemma A.1 (Uniform shrinking margin). *If seven open unit equilateral triangles cover H , then, for some $\varepsilon > 0$, seven closed equilateral triangles of side $1 - \varepsilon$ also cover H .*

A.2. T3-like trace-majorant feasibility. The calculation uses the local wedge and the two orientation charts developed in the exact-trace module below. It is placed here beside the body warning that the output is a closed majorant, not a replacement open triangle.

Lemma A.2 (T3-like translation calculation). *The closed-trace majorant in Proposition 2.7 exists. In the calculation chart, every original T3-like triangle also satisfies $A_i + B_i \leq 1$.*

A.3. Strict-handoff feasibility.

Lemma A.3 (Cyclic strict-handoff selection). *Under the hypotheses of Proposition 2.11, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

A.4. Shared equilateral-enclosure optimization. Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition A.4 (Equilateral enclosure gauge). *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$(18) \quad \Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^2 h_K(R^j n).$$

It is monotone under inclusion, translation invariant, and positively homogeneous. If K is a polygonal hull with at least two distinct points, a minimizing normal can be chosen perpendicular to a hull edge.

A.4.1. Finite caliper criterion. Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$(19) \quad W_F(N) := \sum_{k=0}^2 \max_{z \in F} \langle z, R^k N \rangle.$$

Theorem A.5 (Finite caliper certificate). *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$(20) \quad W_F(\varepsilon J(q - p)) \leq \frac{\sqrt{3}}{2} \|p - q\|.$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length. For $F_x = \{O, A, B, x\}$, every clause is a finite polynomial clause of degree at most four in the coordinates of A, B, x ; at most $12 \cdot 4^3$ clauses are needed.

A.5. Center and V triangle incidence feasibility.

Lemma A.6 (Separation of nonadjacent boundary edges). *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Lemma A.7 (Local wedge calculation). *The incidence conclusions of Lemma 2.4 hold.*

A.6. Exact-trace normalization. Set

$$D_t := 1 - t + t^2, \quad z := \sqrt{D_t}, \quad 0 \leq t \leq 1.$$

After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

In Type I write

$$(21) \quad U = \alpha + y - tx, \quad V = \beta + x - (1 - t)y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\},$$

and in Type II write

$$(22) \quad U = \alpha + (1 - t)x + ty, \quad V = \beta + tx - y, \quad T = \{U \geq 0, V \geq 0, U + V \leq z\}.$$

In both charts

$$(23) \quad V_0 \in \text{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z.$$

The Type-II vertex in the wedge is

$$(24) \quad Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right).$$

Lemma A.8 (Raw $(3, 0)$ trace-normalization calculation). *The translation asserted in Proposition 2.5 exists and preserves both the open and closed traces in H .*

A.7. The local admissible set and radial envelope. Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \text{conv}\{0, au, bv, c(u + v)\}.$$

This is the translate of the body normalization at V_0 , with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$(25) \quad \mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \leq 1\}.$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.9 (Exact local admissible set). *Put*

$$\begin{aligned} s &:= a + b, & d &:= a^2 + ab + b^2, & m &:= \min\{a, b\}, & M &:= \max\{a, b\}, \\ q &:= s^4 - s^2 + ab, \end{aligned}$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2 - 1)c^2 + Mc - M^2, \\ F_S &:= (m^2 - 1)c^2 + (2mM^2 + M)c + M^4 - M^2. \end{aligned}$$

Then

$$(26) \quad \mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S,$$

where all variables lie in $[0, 1]$ and

$$\begin{aligned} \mathcal{A}_L &:= \{d \leq 1, s \leq 1, q \leq 0, F_L \leq 0\}, \\ \mathcal{A}_T &:= \{d \leq 1, s \leq 1, q \geq 0, F_T \leq 0, c \leq 2M\}, \\ \mathcal{A}_S &:= \{d \leq 1, s > 1, F_S \leq 0, c \leq 1/2\}. \end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a, b) := \max\{c \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

Then

$$(27) \quad c_{\max}(a, b) = \begin{cases} 1, & a = b = 0, \\ \frac{C_L(m)}{2M}, & s \leq 1, q \leq 0, (a, b) \neq (0, 0), \\ \frac{1 + \sqrt{4s^2 - 3}}{1 + \sqrt{4s^2 - 3}}, & s \leq 1, q > 0, \\ \frac{M(2mM + 1) - M\sqrt{4d - 3}}{2(1 - m^2)}, & s > 1, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0, 1] : (a, b, c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.8. Midpoint caliper optimization.

Lemma A.10 (Four-point self-midpoint caliper). *If $a + b > 1$, then*

$$\Lambda\left(\text{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.13.

A.9. Signed-center optimization. The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.11 (Signed CE1/CE2 center normal form). *Let T_C be a closed unit equilateral triangle with $O \in \text{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1 - V_0) + a(V_5 - V_0).$$

There are parameters

$$\begin{aligned} 0 < R < 1, \quad W &:= 1 - R, \quad E := \sqrt{1 - RW}, \\ \eta &:= 1 - E, \quad P := E(1 - E), \end{aligned}$$

and nonnegative slacks α, δ , with $k := \eta + \alpha + \delta$, such that

$$\begin{aligned} F_0 &= R + \alpha - a + Wb, \\ (28) \quad F_1 &= Rb + Wa - k, \quad T_C = \{F_0, F_1, F_2 \geq 0\}. \\ F_2 &= W + \delta - b + Ra, \end{aligned}$$

Define

$$(29) \quad \Delta_R := P - \alpha - W\delta, \quad \Delta_L := P - R\alpha - \delta.$$

Then

$$(30) \quad T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta \right], \quad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R},$$

and the possible companion trace is

$$(31) \quad T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha \right] \quad \text{when } \Delta_L > 0, \quad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}.$$

The normalized trace requires $\Delta_R > 0$, and

$$(32) \quad T_C \text{ is CE1} \iff \Delta_L \leq 0, \quad T_C \text{ is CE2} \iff \Delta_L > 0.$$

The six radial exits from O are

$$\begin{aligned} (33) \quad d_0^C &= E - \alpha - \delta, \quad d_1^C = \frac{\delta}{R}, \quad d_2^C = \delta, \\ d_3^C &= \min \left\{ \frac{\alpha}{R}, \frac{\delta}{W} \right\}, \quad d_4^C = \alpha, \quad d_5^C = \frac{\alpha}{W}. \end{aligned}$$

Moreover,

$$(34) \quad T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Lemma A.12 (CE2 total slack). *In the CE2 sign range, with $T := \alpha + \delta$,*

$$(35) \quad T < \frac{\eta}{E}.$$

Corollary A.13 (Common center-boundary formula). *The full center-boundary contribution is*

$$(36) \quad L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}.$$

In CE1,

$$L_{\partial H}(T_C) \leq P \leq \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1 + E} - \frac{E^2(\alpha + \delta)}{RW} < \frac{1}{2}.$$

Lemma A.14 (Adjacent-edge diameter transfer). *For $0 \leq q \leq 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4 - 3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \leq M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \quad 1 - M_0(q) > \frac{q}{2}.$$

APPENDIX B. TRACE-LENGTH OPTIMIZATION

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. Conditional skeleton caps.

Lemma B.1 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.2, one has $L_S(T_C) < 3/2$.*

Lemma B.2 (No-support skeleton calculation). *The hypotheses of Lemma 3.4 imply $L_S(T_i) \leq 2$.*

For the remaining cap, reflect the local directions if necessary so that $0 \leq A_i \leq B_i \leq 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$(37) \quad \begin{aligned} A_i^2 + A_i B_i + B_i^2 &\leq 1, \\ C_i &\leq \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_i B_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\leq 0, \end{aligned}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.9. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.3 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Lemma B.4 (Common skeleton-budget calculation). *Under the hypotheses of Theorem 3.7, the total skeleton trace of the seven triangles is strictly less than 12.*

B.2. Vd1 and Vd2 corner optimization. Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0, 0)$, $V_5 = (1, 0)$, $V_1 = (0, 1)$, $O = (1, 1)$, and $\|(x, y)\|^2 = x^2 + y^2 - xy$.

Proposition B.5 (Vd1/Vd2 corner normal form). *Let T be the closure of an original Vd1 or Vd2 triangle at V_0 , and let A_0, B_0 be its exact reaches toward V_5, V_1 . There is a unique $t > 0$ such that, with $d = \sqrt{t^2 + t + 1}$,*

$$(38) \quad T = \left\{ (x, y) : \begin{aligned} x - (t+1)y &\leq A_0, \\ ty - (t+1)x &\leq tB_0, \\ tx + y &\leq d - A_0 - tB_0 \end{aligned} \right\}.$$

The raw traces, parametrized from the indicated outer vertex, are

$$(39) \quad \begin{array}{l|l} r_0 & 0 \leq s \leq \frac{d - A_0 - tB_0}{t+1} \\ r_1 & \frac{t(1-B_0)}{t+1} \leq s \leq \frac{d - A_0 - tB_0 - 1}{t} \\ r_5 & \frac{1-A_0}{t+1} \leq s \leq d - A_0 - tB_0 - t. \end{array}$$

Intersecting with $[0, 1]$ gives the actual traces, and

$$(40) \quad A_0 + B_0 < \frac{1}{2}.$$

Lemma B.6 (Vd2 adjacent-midpoint cap). *If an original open V triangle has Vd2 closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \emptyset,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

B.3. Boundary contributions.

Lemma B.7 (Boundary-trace table calculation). *All identities and bounds in Theorem 3.8 hold.*

B.4. Common perimeter budgets.

Lemma B.8 (Master perimeter deficit). *Put*

$$\theta := \frac{2}{\sqrt{3}} - 1.$$

Suppose a branch has center contribution $L_C := L_{\partial H}(T_C)$, exactly n supercritical Vd0 triangles, and $m \geq 1$ distinct nonsupercritical V triangles $T_{i_1}, \dots, T_{i_m}$ such that

$$L_{\partial H}(T_{i_j}) < q_j < 1 \quad (1 \leq j \leq m).$$

Suppose also that each of the remaining $6 - n - m$ V triangles contributes at most 1. If

$$(41) \quad L_C + n\theta \leq \sum_{j=1}^m (1 - q_j),$$

then the seven triangles do not cover ∂H .

Lemma B.9 (Small center slacks in the one-Vd branch). *Assume there is exactly one supercritical Vd0 triangle, exactly one Vd1 or Vd2 triangle, and four nonsupercritical Vd0 triangles. Every candidate surviving the perimeter budget is CE2 and satisfies*

$$(42) \quad \alpha + \delta < \frac{1}{24}, \quad \alpha + \delta < \frac{\min\{R, W\}}{6}.$$

Lemma B.10 (CE2 endpoints dominate total slack). *Assume $\Delta_R > 0$ and $\Delta_L > 0$. Put*

$$T := \alpha + \delta, \quad x := \frac{\eta + T}{W}, \quad y := \frac{\eta + T}{R}.$$

Then

$$(43) \quad T < \frac{1}{2} \min\{x, y\}.$$

Lemma B.11 (Perimeter-budget substitutions). *Items (1)–(5) of Corollary 3.9 are impossible.*

APPENDIX C. AREA-LOSS OPTIMIZATION

This appendix evaluates the three area optimizations used in section 4. The body fixes the feasible triangles, the loss functional, and the strict cyclic handoffs; here we introduce coordinates only to solve those problems.

C.1. Wedge chart.

Lemma C.1 (Local wedge reduction). *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \quad W = \{(x, y) : x \geq 0, y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0 , then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x, y) -coordinate area.

C.2. Local square-loss optimization.

Lemma C.2 (Solved local square-loss optimization). *For the feasible family $\mathcal{T}(a, b)$ of section 4, the minimum loss satisfies equations (10) and (11).*

C.3. Cyclic loss optimization. The variables ξ_i, a_i, b_i are the strict handoffs already fixed in section 4; in particular, no endpoint selection is relaxed here.

Lemma C.3 (Solved cyclic area-loss optimization). *The implication in theorem 4.2 hold.*

APPENDIX D. NONZERO-GAP FINITE-ENCLOSURE OPTIMIZATION

This appendix solves the point, radial, disk, and interval calculations used by the nonzero-gap rows of table 2. The geometric witnesses and their case predicates remain in the body.

D.1. Optional B: the four-point two-gap refinement. The former B construction is stronger in point count, but is no longer needed by the active routing. Its separate short-ray calculation is retained for comparison. The active BC construction below takes both boundary points from one selected gap instead.

Lemma D.1 (Two-gap common-pair adapter for a candidate). *In the two-gap path configuration, let an open unit candidate U' contain $X_5(B_5), X_0(1 - A_1), \hat{P}_2, \hat{P}_4$. Then $O \in U'$ and the candidate is CE2 with common midpoint M_0 . For its own signed variables put $p = W - \alpha, q = R - \delta$. In the two-gap path configuration,*

$$A_i > q, \quad B_i > p \quad (1 \leq i \leq 5).$$

Thus every own or permitted neighboring contribution on r_2, r_4 is at most $c_{\max}(p, q)$.

Theorem D.2 (Four-point two-gap enclosure). *Suppose the actual gaps are on $e_{5,0}, e_{0,1}$, the four middle edges are gap-free, and $T_1, \dots, T_5$ are nonsupercritical. Then the fixed set*

$$K_B := \{X_5(B_5), X_0(1 - A_1), \hat{P}_2, \hat{P}_4\}$$

satisfies $\Lambda(K_B) \geq 1$. Skeleton coverage forces $K_B \subset U_C$. There is no disk in this recipe.

D.2. Fixed endpoints and arbitrary candidates. The new enclosure theorems use actual trace endpoints, fixed before an arbitrary candidate is chosen. This subsection supplies the new arguments; the subsequent local capacity and selected-return calculations are retained. The BC tail bound belongs to the original configuration; candidate parameters do not alter its six witness points. The optional B comparison pair likewise belongs to the candidate, while its four witnesses stay fixed.

Lemma D.3 (Total radial endpoint and maximum-to-own-demand). *The assertions of theorem 5.3 hold for arbitrary V types.*

Lemma D.4 (Fixed four-point two-gap enclosure). *The fixed set K_B in theorem D.2 has $\Lambda(K_B) \geq 1$.*

Lemma D.5 (Fixed six-point transverse enclosure). *The set K_{BC} in theorem 5.13 has $\Lambda(K_{BC}) \geq 1$ for arbitrary V types on the nonsupercritical path.*

Lemma D.6 (Fixed four-point supported-rescuer proof). *The geometric statement of theorem 5.17 holds.*

Using the geometric lemma on an actual supported trace. Set the lemma's parameters to $a = A_0$, $\beta = B_5$ and $\varepsilon = 1 - u$. The local T3-like or Vd1 certificate gives

$$A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M_c^{\text{sup}}.$$

The strict-supercritical envelope and path monotonicity give $B_5 \leq B_1 < M_c^{\text{sup}}$. Since $M_c^{\text{sup}} \geq 1/2$, these imply $A_0 \leq \varepsilon$ and $B_5 < \varepsilon/(A_0 + \varepsilon)$. They also imply $A_0 \leq 1 - M_c^{\text{sup}}$ and hence $B_5 < 1 - A_0$. Thus the two boundary points are actual left-gap endpoints, and the other noncentral point is the already forced supported endpoint. This constructs $K_D \subset U_C$ for both gap ranks; the four-point lemma is its universal candidate obstruction.

Scope of the retained calculations below. The local admissible cells, CE1 selected-return calculations, and the two rescuer chart certificates below remain inputs to the fixed-witness proof. The older one-Vd radial-separation estimates and the boundary-tail budget are retained as alternative quantitative results, not as additional active finite-enclosure cases. In particular, an actual-center separation is not itself renamed as an unrestricted enclosure inequality.

D.3. Exact gap endpoints.

Lemma D.7 (Actual open-trace endpoints). *For every actual gap J_i ,*

$$J_i = X_i([B_i, 1 - A_{i+1}]).$$

The formula includes the singleton case $B_i = 1 - A_{i+1}$.

D.4. Common-pair reach adapters.

Lemma D.8 (Solved one-gap common-pair adapter). *The reaches defined from row A in theorem D.17 satisfy its displayed coordinatewise alternatives.*

Lemma D.9 (Two-gap common-pair adapter). *In the B configuration, $T_1, \dots, T_5$ are nonsupercritical by hypothesis. For any candidate containing the fixed K_B , its signed CE2 pair $p = W - \alpha$, $q = R - \delta$ satisfies*

$$A_i > q, \quad B_i > p \quad (1 \leq i \leq 5).$$

Every local contribution on r_2 or r_4 is at most $c_{\max}(p, q)$.

D.5. Vd1 two-chart replacement. The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then invokes the all-nonsupercritical obstruction without assuming preservation of the input gap rank.

Lemma D.10 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \geq \frac{1}{2}, \quad 0 \leq z \leq \frac{1}{2},$$

then

$$y \leq 1 - z.$$

Lemma D.11 (Two-vertex replacement from scalar margins). *Consider two adjacent V roles. Only the five pieces*

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

may be affected by their replacement, and the shared edge is center-free. Let $a, c, B, r \in [0, 1)$ be the boundary reach on $e_{5,0}$, the own-radial reach on r_0 , the boundary reach on $e_{1,2}$, and the radial demand on r_1 needed to overlap the unchanged center interval. If

$$a < \frac{1}{2}, \quad a + c < 1, \quad a + B < 1, \quad r < \max\{1 - a, 1 - B\},$$

there are two open unit equilateral replacements preserving these five skeleton pieces, with actual boundary sums below one and no positive adjacent support. No center-class formula is assumed.

Proposition D.12 (Vd1 input to the two-vertex replacement). *Assume a skeleton cover has a unique Vd1 role T_0 supplying M_1 , a unique supercritical role T_1 , and all other V roles are nonsupercritical Vd0. Suppose $e_{0,1}$ is center-free and no other V role has positive trace on r_0 or r_1 . The scalar replacement applies in CE1 and CE2 and yields a skeleton cover with $N'_+ = 0$.*

D.6. Direct neighboring-ray domination. The capacities keep their variational definitions. The following proof establishes the bound needed by all active witnesses without evaluating the cubic plateau and radical branches of the full function.

Lemma D.13 (Support-cell rotation). *A least equilateral enclosure of a nondegenerate polygon can be chosen with one side containing a polygon edge. For a centered disk together with finitely many points, it suffices to test point–point and point–disk ties, or a disk-only orientation.*

Lemma D.14 (Sharp diagonal neighboring capacity). *For $0 \leq m \leq 1/2$, one has $C_+(m, m) = C_-(m, m) = 1 - m$.*

Lemma D.15 (Type-aware radial calculation). *The points $D_i = (1 - \Gamma_i)V_i$ of theorem 5.4 are missed by every open V role.*

Lemma D.16 (Solved common-pair comparison). *The inequalities in theorem 5.5 hold.*

D.7. Optional alternative A: the complementary-gap disk.

Lemma D.17 (One-gap common-pair adapter). *Suppose $N_+ = 0$ and the unique actual gap is $J_0 = X_0([B_0, 1 - A_1])$. Put $p = A_1$, $q = B_0$. Then $p, q > 0$, $p + q \leq 1$, and $A_i \geq p$, $B_i \geq q$ for all i , without a V -type restriction.*

Put

$$\begin{aligned} c_A &:= c_{\max}(A_1, B_0), & D_i^A &:= (1 - c_A)V_i, \\ G_A &:= \begin{cases} X_0(B_0), & B_0 \leq A_1, \\ X_0(1 - A_1), & A_1 < B_0, \end{cases} & K_A &:= \{D_0^A, \dots, D_5^A, G_A\}. \end{aligned}$$

The disk used in this recipe is exactly

$$\mathcal{D}_A := \left\{ x \in \mathbb{R}^2 : \|x\| \leq \frac{\sqrt{3}}{2} (1 - c_{\max}(A_1, B_0)) \right\}.$$

It is the inscribed disk of the regular hexagon formed by the six radial points. It need not itself be missed pointwise by the V triangles: its containment in U_C follows from convexity of the six forced points.

Theorem D.18 (Complementary-gap finite enclosure). *For $p, q > 0$, $p + q \leq 1$, $c_* = c_{\max}(p, q)$, and $J(p, q) = X_0([q, 1 - p])$, let G be its farther endpoint from O . Then*

$$\Lambda(\mathbb{B}(O, h(1 - c_*)) \cup \{G\}) \geq 1.$$

Consequently the all-nonsupercritical one-gap configuration forces $K_A \subset U_C$ and satisfies $\Lambda(K_A) \geq 1$.

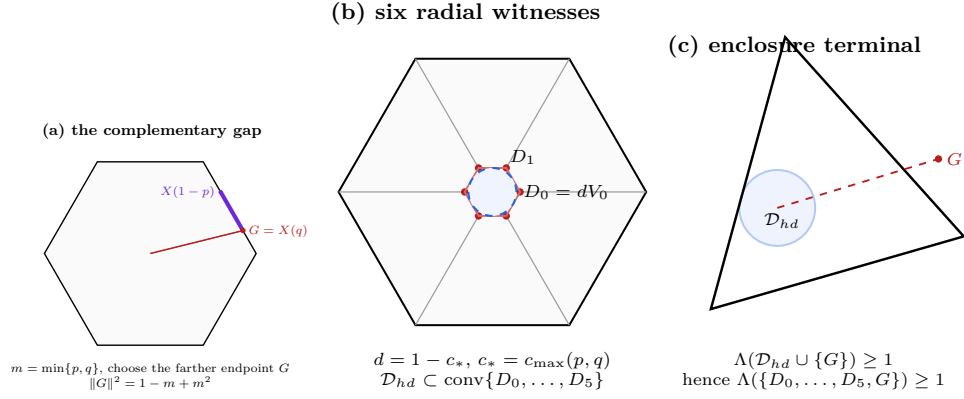
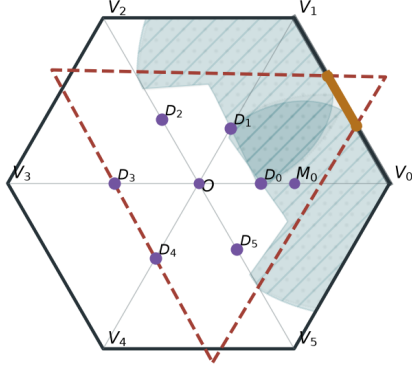
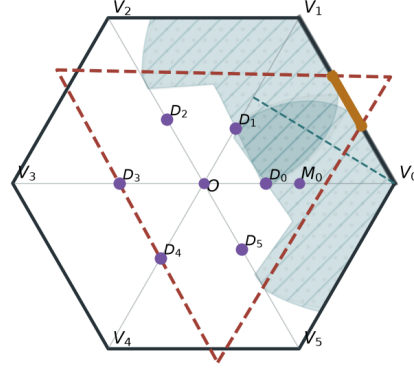


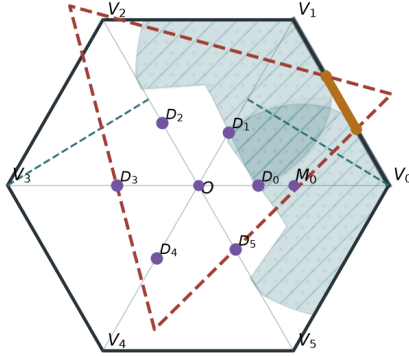
FIGURE 9. The complementary-gap enclosure mechanism. The farther endpoint G of $J(p, q)$ is center-forced. Common-pair domination produces the six center-forced points $D_i = (1 - c_*)V_i$, whose convex hull contains the disk $\mathcal{D}_{h(1-c_*)}$. The disk-plus-point formula then forces enclosing side length at least one. The disk is not an additional witness; it is contained in the convex hull of the six finite radial witnesses.



All Vd0.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 0)$



One T3-like role.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 1)$



Two T3-like roles.
 $N_{\text{gap}} = 1$
 $N_+ = 0$
 $(d, t) = (0, 2)$

FIGURE 10. Case A: sampled complementary-gap data.

D.8. Disk and complementary-gap optimization. Put $\mathcal{D}_\eta := \mathbb{B}(O, \eta)$.

Lemma D.19 (Disk-plus-point formula). *If $\|G\| = \rho$, then*

$$\Lambda(\mathcal{D}_\eta \cup \{G\}) = \begin{cases} 3\eta/h, & \rho \leq 2\eta, \\ \frac{3\eta + \sqrt{3(\rho^2 - \eta^2)}}{2h}, & \rho > 2\eta. \end{cases}$$

(b) $\rho > 2\eta$

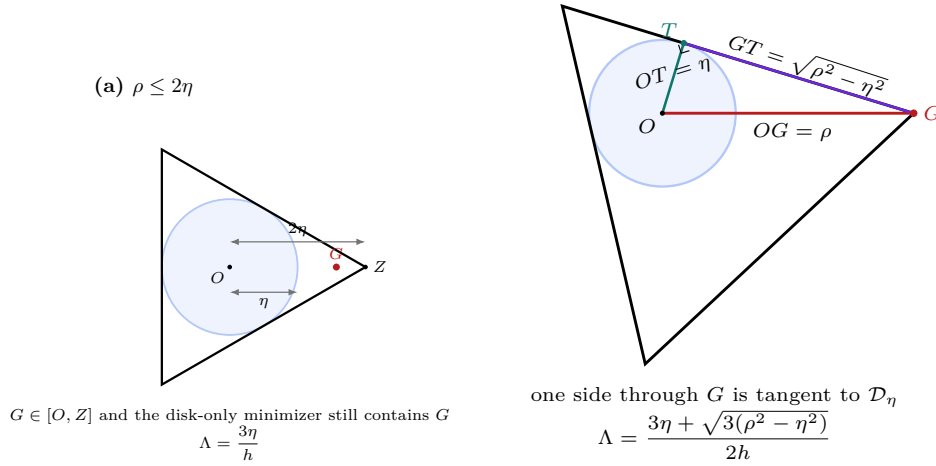


FIGURE 11. The two minimizing configurations in the disk-plus-point formula. For $\rho \leq 2\eta$, the centered equilateral triangle tangent to $\mathcal{D}_\eta$ already contains G . For $\rho > 2\eta$, the point G becomes a vertex of the minimizing triangle and one incident side is tangent to the disk. The tangent length $GT = \sqrt{\rho^2 - \eta^2}$ is the geometric source of the radical term.

Lemma D.20 (Solved complementary-gap optimization). *The set $\mathcal{D}(p, q) \cup J(p, q)$ of theorem D.18 has enclosure number at least one.*

D.9. CE2 short-ray optimization.

Lemma D.21 (Solved CE2 short-ray optimization). *Under the hypotheses of theorem D.2,*

$$c_{\max}(p, q) < 1 - e, \quad e := \min\{\alpha, \delta\},$$

and consequently $\{D_2, D_4\} \not\subseteq T_C$.

D.10. High-radial and selected-chord calculations. The zero-radial diameter envelope is

$$(44) \quad M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2}.$$

For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$(45) \quad \epsilon(e) = \frac{1-e}{2} \left(1 - \sqrt{4(1-e)^2 - 3} \right).$$

It satisfies

$$(46) \quad e < \epsilon(e), \quad \epsilon(e) < \frac{2e}{1-2e},$$

and, for $e \leq 1/8$,

$$(47) \quad \epsilon(e) \leq 2e + 5e^2.$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.22 (Direct high-radial threshold). *Let a nonsupercritical V triangle have incoming boundary reach A and outgoing boundary reach B . If*

$$C \geq 1 - e, \quad A > \epsilon(e),$$

then

$$\boxed{B \leq \epsilon(e)}.$$

For $0 \leq c \leq 1/2$, the strict-supercritical outgoing supremum is

$$M_c^{\sup} = \frac{c + \sqrt{c^2 - 8c + 4}}{2}.$$

It is not attained. Thus an actual strict-supercritical V triangle with own-radial reach at least c has outgoing boundary reach strictly below $M_c^{\sup}$.

Lemma D.23 (Selected-arc chord bounds). *Let $0 < e < 1/2$. On the selected Q_+ component for radial demand $1 - e$, let p be the incoming boundary reach and let q be the following incoming reach forced across a center-free edge. Then*

$$q - p = 1 - \sqrt{1 - x + x^2},$$

where

$$q = e + (1 - e)x, \quad p = e + (1 - e)x - 1 + \sqrt{1 - x + x^2}.$$

The selected arc is increasing and concave. In particular, on every interval used below,

$$q \geq p + \lambda(p - e)$$

for the chord slope λ joining the appropriate selected endpoints. The CE1 applications may take

$$\lambda > 1 - 4e \quad (e = \alpha), \quad \lambda > 1 - 5e \quad (e = \alpha/R).$$

D.11. Exact direct output formulas. The preceding proofs refer to three local inequalities. We derive them here from Proposition A.9. This replaces the former presentation by iterated reach maps.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c , let

$$\mathcal{B}_c(a) = \max\{b : (a, b, c) \text{ is admissible and } a + b \leq 1\}.$$

This is an outgoing capacity for one triangle; it is never composed in the proof.

For $1/2 < c \leq \sqrt{3}/2$, put

$$h(c) = \frac{2c}{1 + \sqrt{16c^2 - 3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2} \left(1 - \sqrt{4c^2 - 3}\right), \quad r(c) = c - \ell(c).$$

Also set

$$Q_-(a, c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a, \\ Q_+(a, c) = \frac{2(1 - a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1 - c^2)(1 - a^2)c^2}}.$$

Proposition D.24 (Four direct high-radial outputs). *For $1/2 < c \leq \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a < h(c), \\ h(c), & a = h(c), \\ Q_-(a, c), & h(c) < a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1 - a, & 0 \leq a \leq 1 - c, \\ Q_+(a, c), & 1 - c < a \leq \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ Q_-(a, c), & r(c) \leq a < c, \\ 1 - a, & c \leq a \leq 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

D.11.1. Vd corner margins. In the exact Vd1/Vd2 corner form, let the boundary reaches be a, b , let $t > 0$, and put $d = \sqrt{t^2 + t + 1}$. The own-radial reach is

$$c_0 = \frac{d - a - tb}{t + 1}.$$

If the role has positive support on the forward adjacent arm, its far endpoint there is

$$u_+ = \frac{d - a - tb - 1}{t}.$$

Since $d < t + 1$,

$$c_0 < 1 - \frac{a + tb}{t + 1} \leq 1 - \min\{a, b\},$$

and

$$u_+ < 1 - b - \frac{a}{t} < 1 - b.$$

Thus lower boundary demands $a \geq A$, $b \geq H$ imply

$$c_0 < 1 - \min\{A, H\}, \quad u_+ < 1 - H.$$

Reflection supplies the backward adjacent-arm statements. These two one-line inequalities are exactly the radial separations used in the one-Vd placement proof.

D.12. CE1 reverse-path calculation. The CE1 part of BC uses a boundary tail floor at T_4 , three middle handoffs, and the transverse radial reaches at T_4, T_3, T_2 . The old five-handoff return is a corollary, not an input to BC.

Proposition D.25 (CE1 tail-input three-transverse obstruction). *Use the signed CE1 center variables*

$$\begin{aligned} 0 < R < 1, \quad w &= 1 - R, \quad E = \sqrt{1 - Rw}, \\ \eta &= 1 - E, \quad P = E(1 - E), \end{aligned}$$

and write

$$A = \alpha, \quad D = \delta.$$

Assume

$$(48) \quad A > 0, \quad D > 0, \quad A + wD < P, \quad D + RA \geq P.$$

Put

$$(49) \quad X = R - D, \quad m = \frac{A}{R}, \quad h_0 = \frac{\eta + A + D}{2R}.$$

Let $T_1, \dots, T_4$ be nonsupercritical V triangles of arbitrary types. All reaches below are actual. Assume

$$A_1 \geq X, \quad B_i + A_{i+1} \geq 1 \quad (1 \leq i \leq 3), \quad B_4 \geq h_0,$$

and

$$(50) \quad C_4 \geq 1 - A, \quad C_3 \geq 1 - m, \quad C_2 \geq 1 - D.$$

These hypotheses are incompatible. Neither a fifth handoff nor a bound on A_0 is part of this local statement.

Corollary D.26 (Retained five-handoff return). *Under the same signed CE1 domain, let T_0 be a V triangle and let $T_1, \dots, T_5$ be nonsupercritical. Assume $A_1 \geq X$, $B_i + A_{i+1} \geq 1$ for $1 \leq i \leq 4$, $B_5 + A_0 \geq 1$, and all three radial hypotheses of equation (50). Then*

$$(51) \quad \boxed{A_0 > 1 - h_0.}$$

D.13. Transverse seven-point calculation.

CE1 reverse path: from the selected-gap tail to the final contradiction

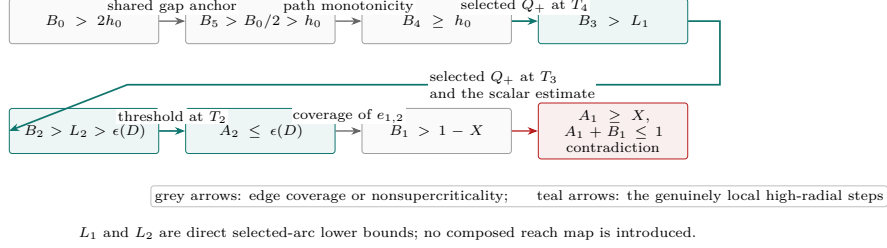


FIGURE 12. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the BC tail input (with $h_0 = Q$), moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

Notation for retained own-endpoint calculations. In the older all-Vd0 specialization below put $\ell = B_0$, $r = 1 - A_1$, $P_i = (1 - C_i)V_i$, and

$$K_{\text{tr}} = \{O, M_0, X_0(\ell), X_0(r), P_2, P_3, P_4\}.$$

These auxiliary own endpoints are distinct from the total endpoints $\hat{P}_i$ of the active BC theorem. Their Vd0-specific forcing is retained only for that specialization; the conditional local return below uses no V-type assumption once its own-radial demands are supplied.

Lemma D.27 (Endpoint upper squeeze). *If an open unit triangle U' contains K_{tr} , use its local signed-center variables and put*

$$X = R - \delta, \quad Q = \frac{\eta + \alpha + \delta}{2R}.$$

Then its incoming reach at V_0 satisfies

$$A_0(U') < 1 - Q.$$

Lemma D.28 (Solved CE2 transverse threshold). *For an open unit equilateral triangle U_C , put $T_C := \overline{U_C}$. Then*

$$N_E(T_C) = 2 \implies K_{\text{tr}} \not\subseteq U_C.$$

D.14. Disk, one-third, and supported-tail optimizations.

Proposition D.29 (Disk-finite-set calipers). *Let $P = \{p_1, \dots, p_m\}$, and let J denote rotation by $\pi/2$. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm \frac{J(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \neq p_k)$$

or

$$\frac{\eta p_i \pm \sqrt{\|p_i\|^2 - \eta^2} Jp_i}{\|p_i\|^2} \quad (\|p_i\| \geq \eta).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Lemma D.30 (Half-edge one-third radial envelope). *If*

$$M \geq \frac{1}{2}, \quad 0 < m \leq M, \quad M + m < 1,$$

then

$$(52) \quad \boxed{c_{\max}(M, m) < 1 - \frac{m}{3}}.$$

Lemma D.31 (Supported endpoint to four-point enclosure). *Suppose T_0 has actual boundary endpoint $a = A_0$ toward V_5 and supported interval $[c, u]$ on r_1 , measured from V_1 toward O . Put $\varepsilon = 1 - u > 0$ and $M = M_c^{\sup}$. Assume*

$$\varepsilon V_1 \in U_C, \quad A_0 + \varepsilon \leq 1, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

$$(53) \quad A_0 \leq 1 - M, \quad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

The first inequality in the last display is a consequence, not an extra hypothesis. If T_1 is uniquely supercritical with $C_1 \geq c$, and the four center-free edges from $e_{1,2}$ through $e_{4,5}$ join nonsupercritical T_2, T_3, T_4, T_5 , then the skeleton is not covered. The conclusion applies to both possible nonzero gap ranks.

D.15. T3-like supported-tail adapter.

Lemma D.32 (Solved T3-like supported-tail adapter). *The exact hypotheses of theorem 5.19 imply the rescuer-tail hypotheses in theorem D.31.*

D.16. Alternative one-Vd radial-separation calculations. The calculations in this subsection are retained quantitative alternatives. They assume the original reduced CE2 one-Vd placement, its boundary-cover and strict Vd-cap hypotheses, and all the corresponding residual selectors. They are not deductions from the bare CE2 signed domain, and are not used by the active B/C fixed-point route.

For radial separation along r_i , define

$$d_i^C := \max\{x \in [0, 1] : xV_i \in T_C\},$$

$$c_{\text{loc}} := \sup\{s \in [0, 1] : V_i + s(O - V_i) \in U_{i-1} \cup U_i \cup U_{i+1}\}.$$

Then

$$T_C \cap r_i = \{xV_i : 0 \leq x \leq d_i^C\},$$

$$(U_{i-1} \cup U_i \cup U_{i+1}) \cap r_i \subseteq \{xV_i : 1 - c_{\text{loc}} \leq x \leq 1\}.$$

Thus the exact strict inequality

$$c_{\text{loc}} < 1 - d_i^C$$

leaves the nonempty interval

$$\{xV_i : d_i^C < x < 1 - c_{\text{loc}}\} \subseteq S \setminus \left(U_C \cup \bigcup_{j=0}^5 U_j \right).$$

For the nonadjacent adapter, if $I \subseteq [0, 1]$ is empty or a closed interval, define

$$\mathcal{R}_I(p) := 1 - \sup\{x \in [0, 1] : [0, x] \subseteq [0, p] \cup I\}.$$

Put

$$I_R := \{x \in [0, 1] : X_0(x) \in T_C\}, \quad I_L := \{x \in [0, 1] : X_5(1 - x) \in T_C\},$$

$$\rho_R := \mathcal{R}_{I_R}(B_0), \quad \rho_L := \mathcal{R}_{I_L}(A_0), \quad \rho := \min\{\rho_R, \rho_L\}, \quad D_\tau := \rho V_\tau.$$

Appendix D proves the exact separation inequalities for both adapters.

Lemma D.33 (Solved adjacent Vd separation). *Under the adjacent placement*

$$\sigma = 0, \quad \tau \in \{1, 5\},$$

the radial-separation inequality $c_{\text{loc}} < 1 - d_2^C$ holds on the routed arm r_2 .

Lemma D.34 (Solved nonadjacent Vd separation). *Under the nonadjacent placement*

$$\sigma = 0, \quad \tau \in \{2, 3, 4\},$$

the point $D_\tau = \min\{\rho_R, \rho_L\}V_\tau$ lies beyond both the C-triangle exit and the actual Vd radial endpoint.

Lemma D.35 (Solved Vd1 supported-tail adapter). *When the unique Vd1 role is based at T_0 , its actual corner data imply the two strict-supercritical envelope inequalities required by theorem D.31.*

APPENDIX E. ZERO-GAP NINE-POINT OPTIMIZATION

This appendix supplies exact frontier and radial calculations, point exclusions, Newton reduction, and the four-contact enclosure proof for the type-independent zero-gap obstruction. The radial step reuses the common C_+, C_- forcing interface; the three asymmetric witnesses and the exact certificate are unchanged. It uses the public parameters and witness $K_{\text{wit}}(a, b)$ from section 5.6. The symbols A, B, C introduced below are appendix-only Newton inner points required by the exact certificate; they do not replace the public frontier points Q_-, Q_0, Q_+ .

The finite caliper criterion used here is the shared result theorem A.5 in Appendix A. It is not rederived in this appendix.

E.1. Corner-clipped AB -unions. Apply the rigid motion that sends the supercritical corner V_4 to the local origin and its two incident edge directions to e_A, e_B . Within this calculation O, A, B, C denote the local origin, anchors, and corner cone; the feasible union keeps the public symbol $\mathcal{R}_{AB}(a, b)$.

Let e_A, e_B be unit vectors with angle 120° , put

$$O = 0, \quad A = ae_A, \quad B = be_B, \quad C = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$(54) \quad \mathcal{R}_{AB}(a, b) = C \cap \bigcup \{T : T \text{ is a closed unit equilateral triangle and } O, A, B \in T\}.$$

The finite description in Theorem A.5 applies to the four-point set $\{O, A, B, x\}$ and makes membership in (54) an exact finite semialgebraic condition of degree at most four.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB -union). *Assume*

$$a + b > 1, \quad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\begin{aligned}\alpha &= \frac{h(a + 2b - aD)}{2\rho}, & \beta &= \frac{h(a - b + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-a + b + (a + b)D)}{2\rho}, & \delta &= \frac{h(2a + b - bD)}{2\rho}.\end{aligned}$$

Then all four coefficients are positive and

$$(55) \quad \mathcal{R}_{AB}(a, b) = (C \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (C \cap \mathcal{D}_B \cap \mathcal{H}_B),$$

where

$$\mathcal{D}_A = \{(u, v) : (u - a)^2 + v^2 - (u - a)v \leq 1\},$$

$$\mathcal{D}_B = \{(u, v) : u^2 + (v - b)^2 - u(v - b) \leq 1\},$$

$$\mathcal{H}_A = \{(u, v) : \alpha(u - a) + \beta v \leq 0\},$$

$$\mathcal{H}_B = \{(u, v) : \gamma u + \delta(v - b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the *A-circle*, the *A-line*, the *B-line*, and the *B-circle*.

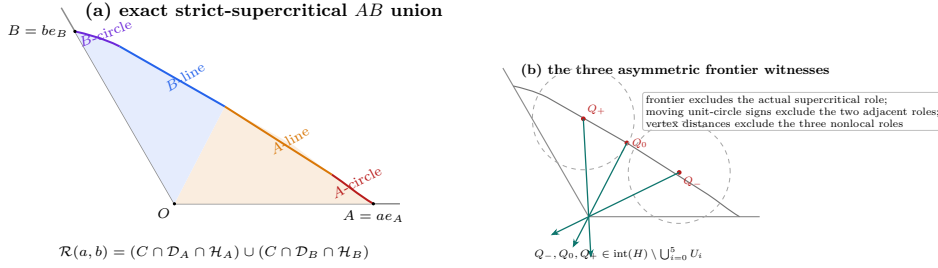


FIGURE 13. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an *A-circle* arc, an *A-line* segment, a *B-line* segment, and a *B-circle* arc. The points Q_- , Q_0 , Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

E.2. Exact-one handoffs and common reach bounds.

Lemma E.2 (Exact-one handoff chain). *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.11 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \quad b = \xi_4, \quad p = 1 - b, \quad q = 1 - a,$$

one has

$$(56) \quad 0 < a, b < 1, \quad a + b > 1, \quad a^2 + ab + b^2 < 1,$$

and every selected V triangle obeys

$$a_i \geq p, \quad b_i \geq q.$$

E.3. Six directly forced radial points. Put $h = \sqrt{3}/2$ and

$$s = p + q, \quad m = \min(p, q), \quad M = \max(p, q), \quad \chi = s^4 - s^2 + pq.$$

The strict domain (56) gives

$$(57) \quad pq > (1-s)(2-s), \quad m > 1-s, \quad s > 1-m.$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.9. On the present subcritical domain, equation (27) becomes

$$(58) \quad c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \frac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s .

Lemma E.3 (Direct radial forcing). *One has*

$$(59) \quad 0 < c_* < 1, \quad c_* > 1 - m.$$

Assume that $U_C, U_0, \dots, U_5$ cover H , with no restriction on the normalized V types. Then, for every i ,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$(60) \quad \mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \quad \eta = h(1 - c_*).$$

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = V_4 + u(V_5 - V_4) + v(V_3 - V_4), \quad \mathbf{q}(u, v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a . Accordingly, set

$$\begin{aligned} \alpha &= \frac{h(b + 2a - bD)}{2\rho}, & \beta &= \frac{h(b - a + (a + b)D)}{2\rho}, \\ \gamma &= \frac{h(-b + a + (a + b)D)}{2\rho}, & \delta &= \frac{h(2b + a - aD)}{2\rho}, \end{aligned}$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The two line pieces of Theorem E.1 are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \quad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D + 3))$ in

$$(61) \quad J = \left(\frac{2b + a - aD}{D + 3}, \frac{b + 2a - bD}{D + 3} \right).$$

Put

$$E_- = (1 - b, 2 - b), \quad E_+ = (2 - a, 1 - a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1 + t, t), \quad 1 - a \leq t \leq b, \quad \text{and} \quad Z_3(y) = (y, 1 + y), \quad 1 - b \leq y \leq a.$$

The restrictions of the two unit-circle equations are

$$(62) \quad \begin{aligned} g_-(\lambda) &= \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2, \\ g_+(\mu) &= \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \end{aligned}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\begin{aligned} \lambda_- &= 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6}, \\ \mu_+ &= 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}. \end{aligned}$$

Lemma E.4 (Moving adjacent-triangle signs). *Throughout (56),*

$$(63) \quad \lambda_* < \lambda_- < h^{-1}, \quad \mu_* < \mu_+ < h^{-1}.$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P, t) = \mathbf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1 - a, b]$,

$$(64) \quad F(L_+(\mu_+), t) \geq 0, \quad F(J, t) > 0, \quad F(L_-(\lambda_-), t) > 0.$$

Under the reflection $(u, v, a, b) \leftrightarrow (v, u, b, a)$, the analogous signs hold for every $y \in [1 - b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Define the local and global witnesses by

$$(65) \quad Q_-^{\text{loc}} = L_-(\lambda_-), \quad Q_0^{\text{loc}} = J, \quad Q_+^{\text{loc}} = L_+(\mu_+),$$

and by the displayed conversion from (u, v) to the plane.

Lemma E.5 (Direct asymmetric forcing). *Assume that $U_C, U_0, \dots, U_5$ cover H . Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^5 U_i,$$

and consequently all three witnesses belong to U_C .

The nine directly forced points are $D_0, \dots, D_5, Q_-, Q_0, Q_+$. Under a putative cover, Lemmas E.3 and E.5 therefore force the compact set

$$(66) \quad K_{\text{wit}}(a, b) = \mathcal{D}_\eta \cup \{Q_-, Q_0, Q_+\} \subset U_C.$$

E.4. Newton inner reduction. For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D+3)}.$$

Let g_-, g_+ be the rescaled forms of (62), and take one Newton step at the junction:

$$\hat{\xi}_- = \xi_* - \frac{g_-(\xi_*)}{g'_-(\xi_*)}, \quad \hat{\zeta}_+ = \zeta_* - \frac{g_+(\zeta_*)}{g'_+(\zeta_*)}.$$

Define, in local coordinates,

$$\begin{aligned} A &= \left(b - \frac{3}{4}\widehat{\beta\xi}_-, \frac{3}{4}\widehat{\alpha\xi}_- \right), \\ B &= J, \\ C &= \left(\frac{3}{4}\widehat{\delta\zeta}_+, a - \frac{3}{4}\widehat{\gamma\zeta}_+ \right), \end{aligned}$$

and convert them to global vectors.

Lemma E.6 (Newton inner reduction). *The points above have coordinates rational in a, b, D and satisfy*

$$A \in (Q_0, Q_-), \quad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \text{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

obeys $\widehat{K} \subseteq \text{conv}(K_{\text{wit}})$.

E.5. Ordered convex chain and supporting-line bounds.

Proposition E.7 (Four-contact witness geometry). *Assume (56) and $c_* > 2/3$. The rays A, B, C occur in that order in one open 120° cone, and*

$$\text{cross}(B - A, C - B) > 0, \quad \|A\|, \|C\| > \eta.$$

Moreover the outward distances of the lines AB, BC satisfy

$$(67) \quad d_{AB} := \frac{\text{cross}(A, B)}{\|B - A\|} \geq h - 2\eta, \quad d_{BC} := \frac{\text{cross}(B, C)}{\|C - B\|} \geq h - 2\eta.$$

E.6. Newton reduction in the body notation. The containment established by theorem E.6 is recorded under the public routing label

$$(68) \quad \widehat{K} = \text{conv}(\mathbb{B}(O, \eta) \cup \{A, B, C\}) \subseteq \text{conv}(K_{\text{wit}}(a, b)), \quad \Lambda(K_{\text{wit}}(a, b)) \geq \Lambda(\widehat{K}).$$

E.7. Four supporting contacts and paired radius transfer. Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For the inner witness define

$$H(n) = \max\{\eta, \langle A, n \rangle, \langle B, n \rangle, \langle C, n \rangle\}, \quad \Psi(n) = \sum_{j=0}^2 H(R^j n).$$

The support formula gives $\Lambda(\widehat{K}) = h^{-1} \min_{\|n\|=1} \Psi(n)$. The minimum can be located without introducing another convex body.

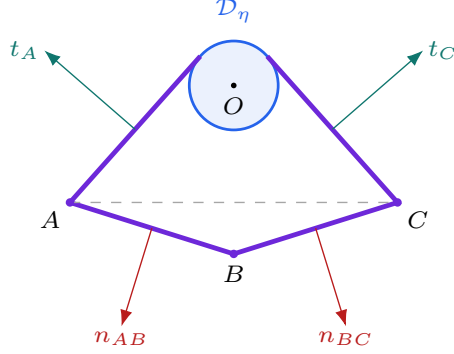
Lemma E.8 (Four-contact enclosure formula). *For the Newton inner witness in the range $c_* > 2/3$, put*

$$\begin{aligned} n_{AB} &= -\frac{J(B - A)}{\|B - A\|}, & n_{BC} &= -\frac{J(C - B)}{\|C - B\|}, \\ t_A &= \frac{\eta A - \sqrt{\|A\|^2 - \eta^2} JA}{\|A\|^2}, & t_C &= \frac{\eta C + \sqrt{\|C\|^2 - \eta^2} JC}{\|C\|^2}. \end{aligned}$$

Then

$$(69) \quad \Lambda(\widehat{K}) = \frac{1}{h} \min\{\Psi(n_{AB}), \Psi(n_{BC}), \Psi(t_A), \Psi(t_C)\}.$$

(a) the four exposed contacts



(b) four support sums

Two consecutive-point contacts:

$$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h,$$

$$\Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

Two exposed disk tangencies:

$$P_A(\eta), P_C(\eta) \geq 0$$

$$\implies \Psi(t_A), \Psi(t_C) \geq h.$$

$$\text{Hence } \Lambda(\hat{K}) \geq 1.$$

FIGURE 14. The four-contact terminal for the zero-gap witness. The supporting lines through AB and BC and the two outer disk tangencies exhaust the active changes of support source. The dashed segment AC is not exposed: B and the disk center lie on opposite sides. The drawing is schematic; the exact witness geometry and the paired tangent residuals are proved in the text. No auxiliary Minkowski sum or cap covering is needed.

At the first two contacts the disk supplies the other two supports, so

$$(70) \quad \Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h, \quad \Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$$

For the two tangencies, put

$$\Delta = \text{cross}(C, RA) > 0, \quad \nu = \langle C, RA \rangle, \quad N_X = \|X\|^2 \quad (X = A, C).$$

Choosing C as a support source at Rt_A and A at R^2t_C , with the disk in the other directions, gives

$$(71) \quad \Psi(t_A) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_A - \eta^2}}{N_A}, \quad \Psi(t_C) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_C - \eta^2}}{N_C}.$$

Consequently the only remaining targets are the paired residual signs

$$(72) \quad P_X(\eta) := (N_X - \eta^2)\Delta^2 - ((h - 2\eta)N_X - \eta\nu)^2 \geq 0 \quad (X = A, C).$$

Indeed $P_X(\eta) \geq 0$ gives $\Delta\sqrt{N_X - \eta^2} \geq |(h - 2\eta)N_X - \eta\nu|$, which implies the required lower bound even when the expression inside the absolute value is negative. No sign is lost by squaring.

The exact certificate uses a smaller rationally specified radius. The following elementary transfer is the reason those existing signs suffice for the actual-radius tangencies.

Lemma E.9 (Simultaneous radius transfer). *Let X, Y be vectors, $d \geq 0$, and $0 \leq e_0 \leq 1/3$. If*

$$\|(1 - 2e_0)X - e_0Y\| \leq d, \quad \|(1 - 2e_0)Y - e_0X\| \leq d,$$

then both inequalities hold at every $e \in [e_0, 1/3]$.

Lemma E.10 (Solved witness-enclosure optimization). *For every parameter pair in equation (14),*

$$\Lambda(K_{\text{wit}}(a, b)) \geq 1.$$

APPENDIX F. EXACT DISK-TANGENT RESIDUAL CERTIFICATE

The two disk-tangent contact bounds in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms; the repository provenance manifests identify the accompanying code and sparse integer data. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. Certificate manifest. The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA-256 digest

dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. Rational radial upper envelopes. Retain the notation of Section E. Thus

$$p = 1 - b, \quad q = 1 - a, \quad s = p + q, \quad m = \min\{p, q\}, \quad M = \max\{p, q\},$$

$$\chi = s^4 - s^2 + mM, \quad h = \frac{\sqrt{3}}{2}, \quad c_* = c_{\max}(p, q).$$

The strict witness domain gives

$$(73) \quad mM > (1 - s)(2 - s), \quad m > 1 - s, \quad s > \frac{2}{3}.$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \quad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s + 1)(2s - 1)^2 > 0.$$

Define

$$(74) \quad \bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \leq 0), \quad \bar{c}_T = s - \frac{\chi}{1 - m} \quad (\chi \geq 0).$$

Lemma F.1 (Branchwise radial upper bounds). *On the corresponding radial cells,*

$$c_* \leq \bar{c}_L < 1, \quad c_* \leq \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_$.*

Reflect so that $z = a - b \geq 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \quad m = \frac{1 - U - z}{2}, \quad D^2 = z^2 + 6U + 3U^2 =: K(U, z),$$

and

$$\Xi(U, z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$

$$G(U, z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$(75) \quad \bar{c}_L = 1 - U - \frac{\Xi(U, z)}{2G(U, z)}, \quad \bar{c}_T = 1 - U - \frac{\Xi(U, z)}{2(1 + U + z)}.$$

F.3. Gram reduction and the geometric implication. On the active cell let $\bar{c}$ denote the corresponding expression in (75), and put

$$\bar{\eta} = h(1 - \bar{c}), \quad e = 1 - \bar{c}, \quad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \leq \eta = h(1 - c_*)$. The algebra below is evaluated at $\bar{\eta}$; theorem E.9 will transfer both residual signs to the actual radius η .

Represent a global vector as (x, hy) and put $C' = R^{-1}C$. Define

$$N_A = \|A\|^2, \quad N_C = \|C\|^2, \quad \nu = \langle A, C' \rangle, \\ \Delta_0 = x_{C'}y_A - y_{C'}x_A > 0.$$

Thus $h\Delta_0 = \text{cross}(C', A) = \text{cross}(C, RA) > 0$. The authenticated code uses the opposite signed determinant, but every occurrence is squared; its integer polynomials and digest are unchanged. The Gram identity is

$$(76) \quad h^2\Delta_0^2 = N_A N_C - \nu^2.$$

Define

$$(77) \quad Q_A = \Delta_0^2 - \|tA - eC'\|^2, \quad Q_C = \Delta_0^2 - \|tC' - eA\|^2.$$

Lemma F.2 (Residual signs imply the actual-radius tangent bounds). *If $Q_A, Q_C \geq 0$, then the two actual-radius residuals $P_A(\eta), P_C(\eta)$ in equation (72) are nonnegative. Consequently both tangent contact sums in equation (71) are at least h .*

F.4. Eight exact integer-polynomial signs. The Newton witnesses A, C are rational functions of U, z, D . Substituting (75) into (77), clearing denominators, and reducing by

$$D^2 - K(U, z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$(78) \quad \text{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U, z) + DB_{B,X}(U, z)\},$$

where all eight core polynomials lie in $\mathbb{Z}[U, z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U, z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C , both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. **Three global positive-basis certificates.** Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \quad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$

$$U_{\Delta} = 1 - \frac{\sqrt{3}}{2}, \quad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U -range	z^2 -range
T	$0 \leq U \leq U_{\Delta}$	$0 \leq z^2 \leq \mathcal{P}(U)$
L_0	$0 \leq U \leq U_{\Delta}$	$\mathcal{P}(U) \leq z^2 \leq \mathcal{G}(U)$
L_1	$U_{\Delta} \leq U \leq U_{\max}$	$0 \leq z^2 \leq \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \quad z = y \frac{9-x^4}{8x^2}, \quad 1 \leq x \leq \sqrt{3}, \quad 0 \leq y \leq 1,$$

and then $x = 1 + (\sqrt{3} - 1)r$. This maps $[0, 1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x , the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L -cell polynomial write

$$F(U, z) = E_F(U, w) + zO_F(U, w), \quad w = z^2,$$

and

$$R_F(U, w) = E_F(U, w)^2 - wO_F(U, w)^2.$$

The implications $E_F \geq 0$ and $R_F \geq 0$ give $E_F \geq \sqrt{w}|O_F|$, hence $F \geq 0$ for $z \geq 0$. Use the exact charts

$$U = U_{\Delta}r, \quad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0 , and

$$U = U_{\Delta} + (U_{\max} - U_{\Delta})r, \quad w = s\mathcal{G}(U)$$

for L_1 . The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. If $P(r, s) = \sum a_{ij}r^i s^j$ has bidegree (m, n) , its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \leq k} \sum_{j \leq \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact tangent-residual certificate). *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in (78) are nonnegative. Hence $Q_A, Q_C \geq 0$, and the two actual-radius tangent contact bounds in equation (71) hold.*

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
verify_mixed_overlap_core_derivation.py
verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.